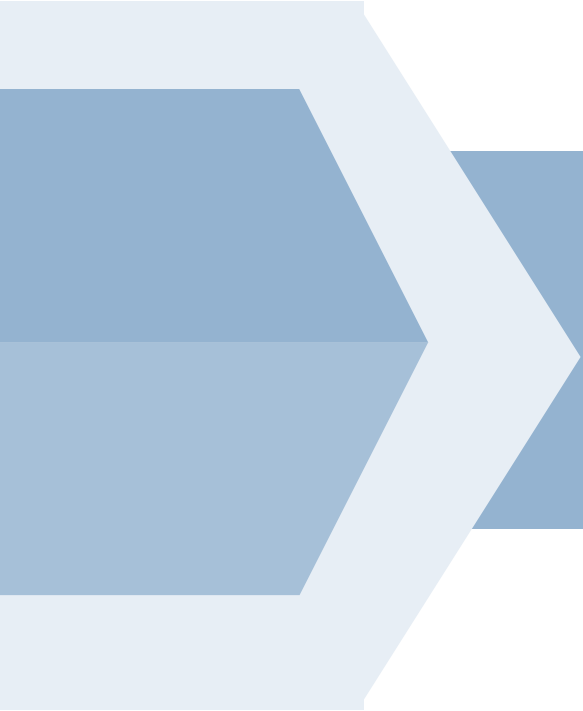


Foundation Models and Generative Artificial Intelligence *ARIN 5203*

Instructor: Dr. Qifeng Chen

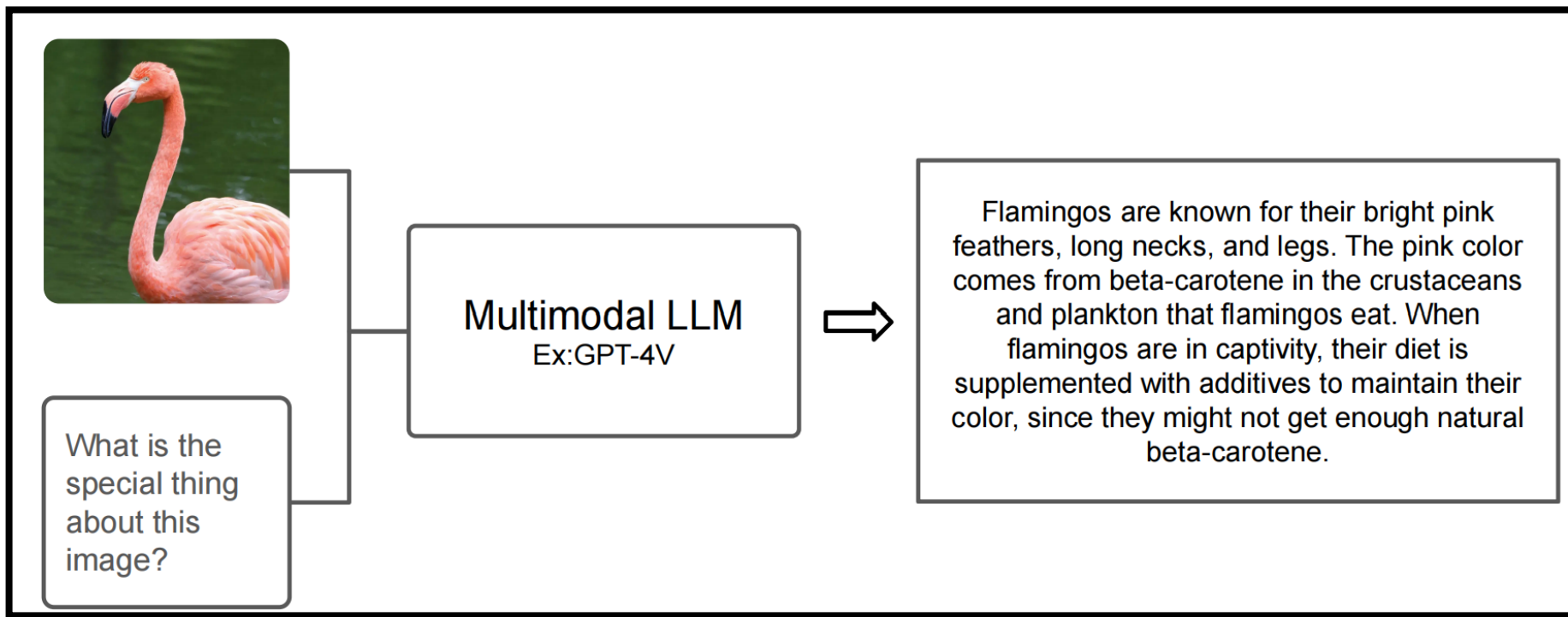


Multimodal LLMs

- **Preliminary**
- **Challenges & Future Work**
- **Representative Models**

■ MLLM Overview / Multimodal Tasks / General MLLM Architecture

Multimodal Large Language Models(Multimodal LLMs)
takes images  and texts  and produce texts. 



■ MLLM Overview / Multimodal Tasks / General MLLM Architecture

Multimodal Datasets — Standard image-text pairs

	T_w: "a week spent at our rented beach house in Sandbridge" T_s: "an outdoor walkway on a grass covered hill"		T_w: "that's what a sign says over the door" T_s: "the car is driving past a small old building"		T_w: "hand held through the glass in my front bedroom window" T_s: "a moon against the night sky with a black background"
	T_w: "stunning sky over walney island, lake district, july 2009" T_s: "an outdoor walkway on a grass covered hill"		T_w: "living in my little white house" T_s: "a tiny white flower with a bee in it"		T_w: "the pink rock from below" T_s: "some colorful trees that are on a hill in the mountains"

■ MLLM Overview / Multimodal Tasks / General MLLM Architecture

Multimodal Datasets — Visual Instruction Data

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation

Question: What type of vehicle is featured in the image?

Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

Response type 3: complex reasoning

Question: What challenges do these people face?

Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

■ MLLM Overview / Multimodal Tasks / General MLLM Architecture

Multimodal Datasets ——— Images interleaved with texts

HTML



Tobin & Duff getting ready to go!
This weekend the hubs and his brother are doing the Ride to Conquer Cancer!
It is a 2 day cycling event, that takes them on a 200+km ride along the front ranges of our Rocky Mountains!...



Over 2200 cyclists this year!

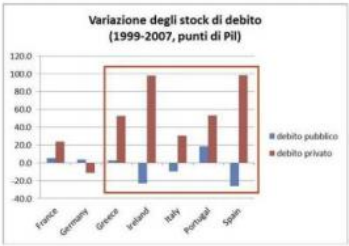


Every year he rides for someone we know, touched by cancer.
This year he rides for his Grandma, my Grandma, Auntie Deb, Art L., Dave L.
2026
May you never have to be touched by cancer.
Blessings & Good Health,

PDF



Please note the balance of euro states trade: Italy in surplus, Germany deficit. Another world indeed. Consider that Italian public debt was not different than today, but it was not a matter of concern, the shock did reflect on the exchange rate, no one was thinking about selling government bonds under par... Please check,



the following graph: red is gdp% shift in private debt and blue public debt from 1999 to 2007, so much for another of the myths of this crisis, the one that says that the fault is of the public debt 2010...

■ MLLM Overview / Multimodal Tasks / General MLLM Architecture

Tasks Introduction

Image captioning



Model

A man drives along a sunlit road, his attention captured by the path ahead, reflecting the everyday adventure of travel and the comfort of a familiar drive."

Visual reasoning



What kind of research is the scientist doing based on the flowchart in the image?

Model

A scientist studies a detailed flowchart in the lab, showcasing her expertise in data analysis.

Visual Dialogue



Human: How many dogs and cats that are in the image?

Assistant: *There are three cats and three dogs.*

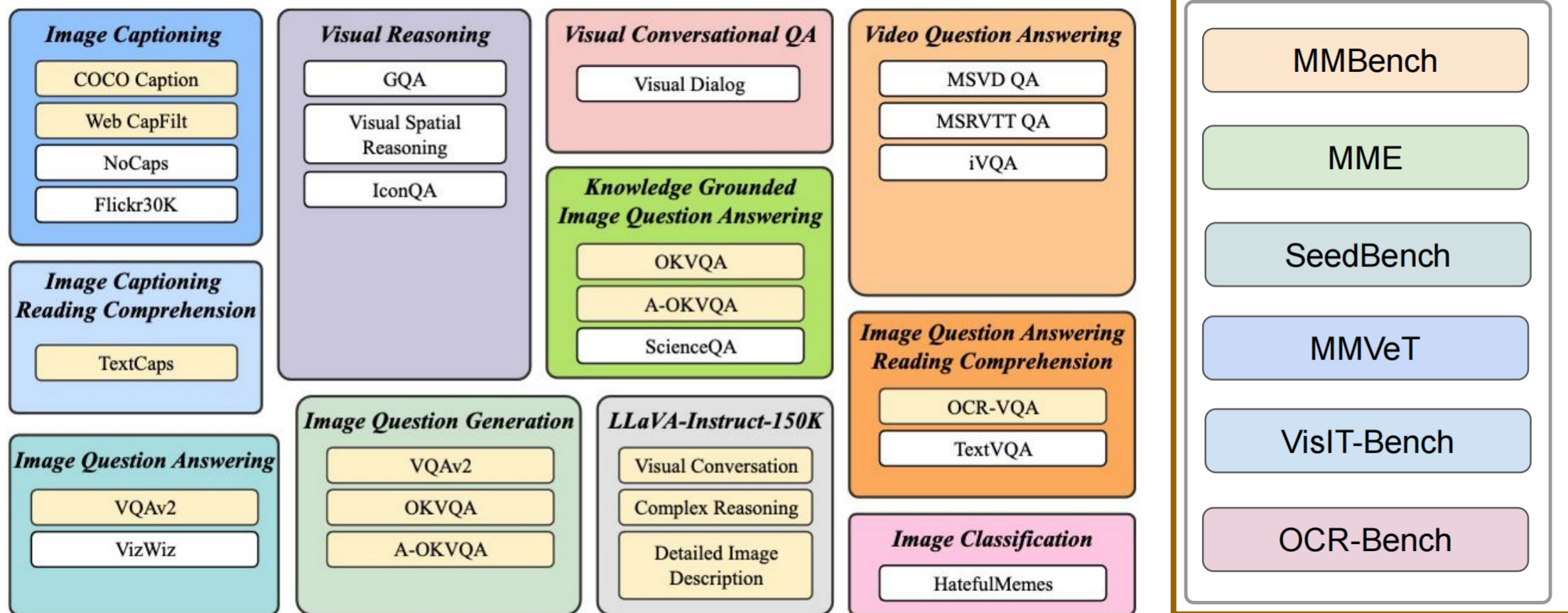
Human: What is the kid in the image holding?

Assistant: *The kid is holding a baby cat.*

■ MLLM Overview / **Multimodal Tasks** / General MLLM Architecture

Benchmarks Introduction

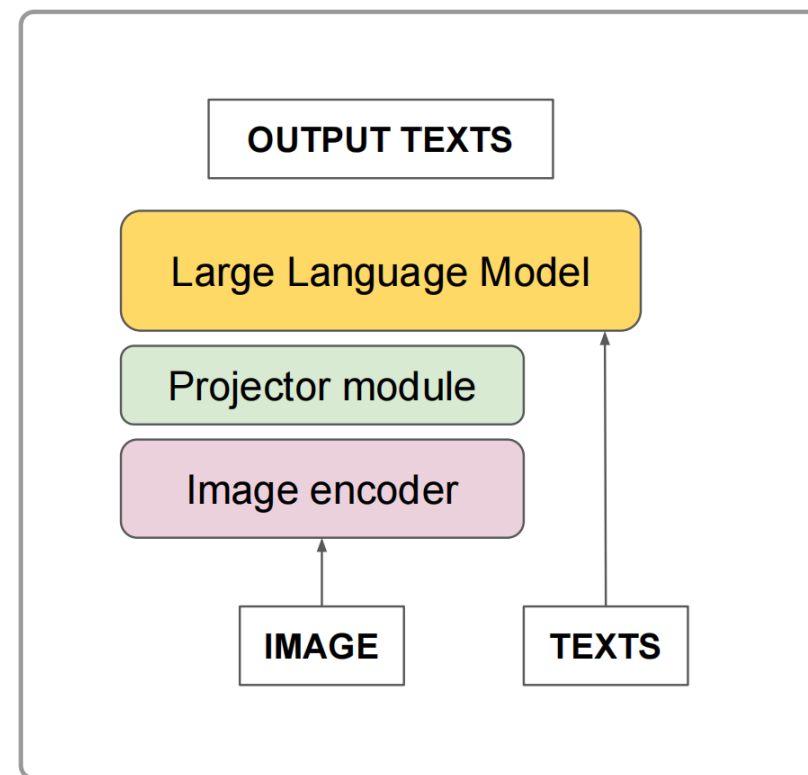
New Multimodal Benchmarks



■ MLLM Overview / Multimodal Tasks / General MLLM Architecture

Most Multimodal LLMs are made of 3 core elements:

- **Image encoder** that takes image and produces image embeddings
- **Projector** or connector module that maps image embeddings to LLMs
- **LLM (often a decoder type)** that generate texts



■ MLLM Overview / Multimodal Tasks / General MLLM Architecture

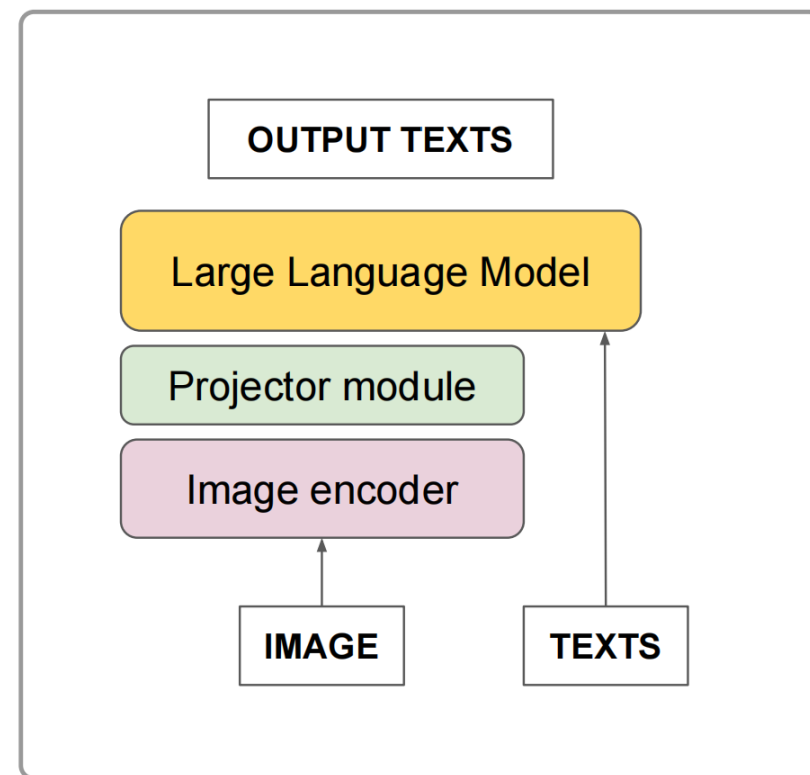
Training Task

Language modelling objective conditioned on image and text tokens.

$$\text{Loss} = - \sum_{i=1}^M \log p(t_{i+1} | v_1, v_2, \dots, v_N, t_1, t_2, \dots, t_i)$$

Training Data

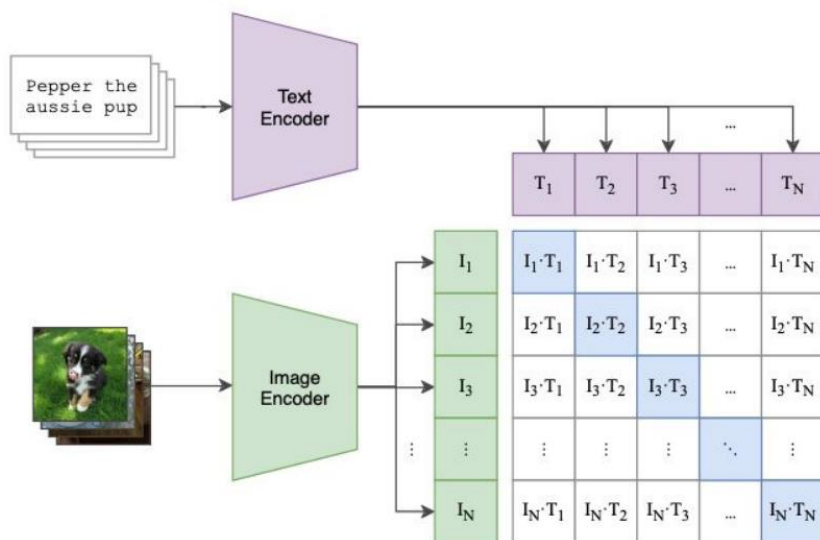
Image text pairs, visual instruction data, images interleaved with texts.



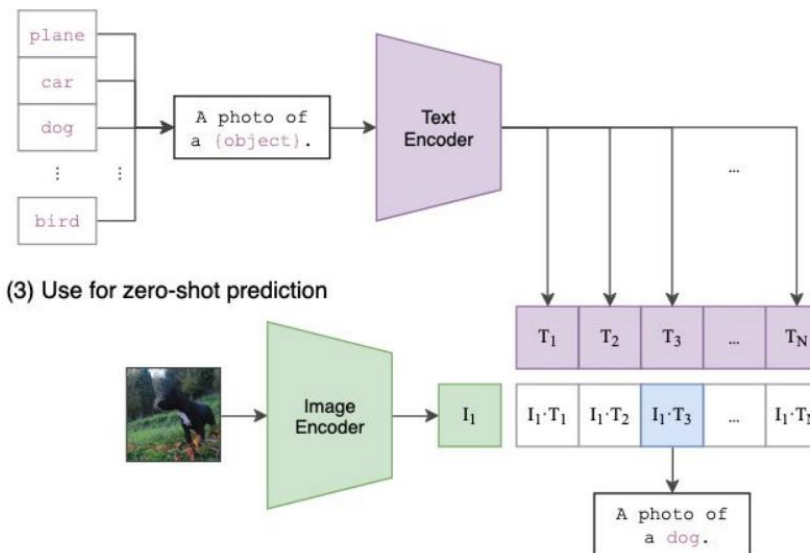
■ CLIP / Flamingo / LLaVA / Others

- CLIP is the one of **the earliest visual language models** that explored image-text pretraining at a very large scale.
- CLIP jointly train an image and text encoder to learn the correct pairings of images and texts using **contrastive loss**.
- During test time, text encoder can be discarded and image encoder can be used for zero-shot classification and image-text retrieval tasks.

(1) Contrastive pre-training



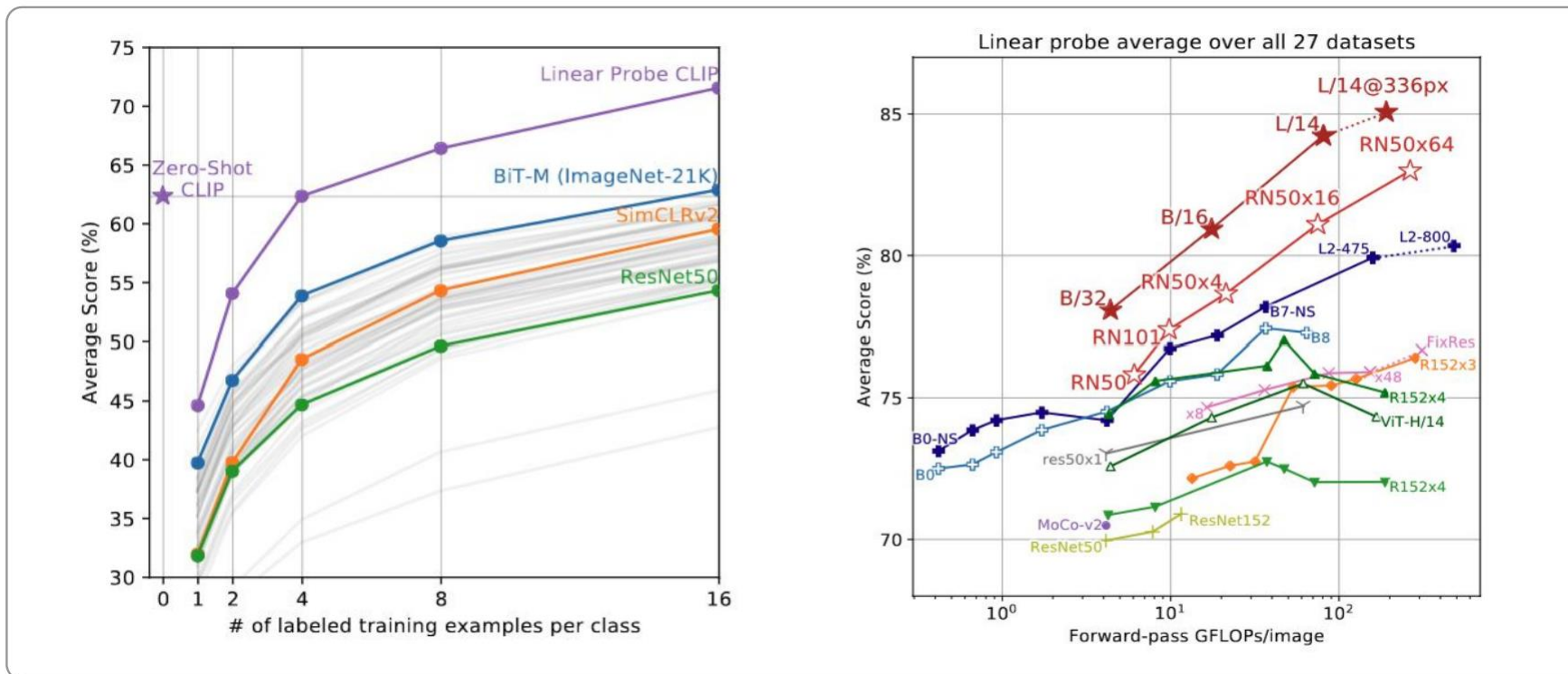
(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

Trained on 400M
image-text pairs
collected from the
internet.

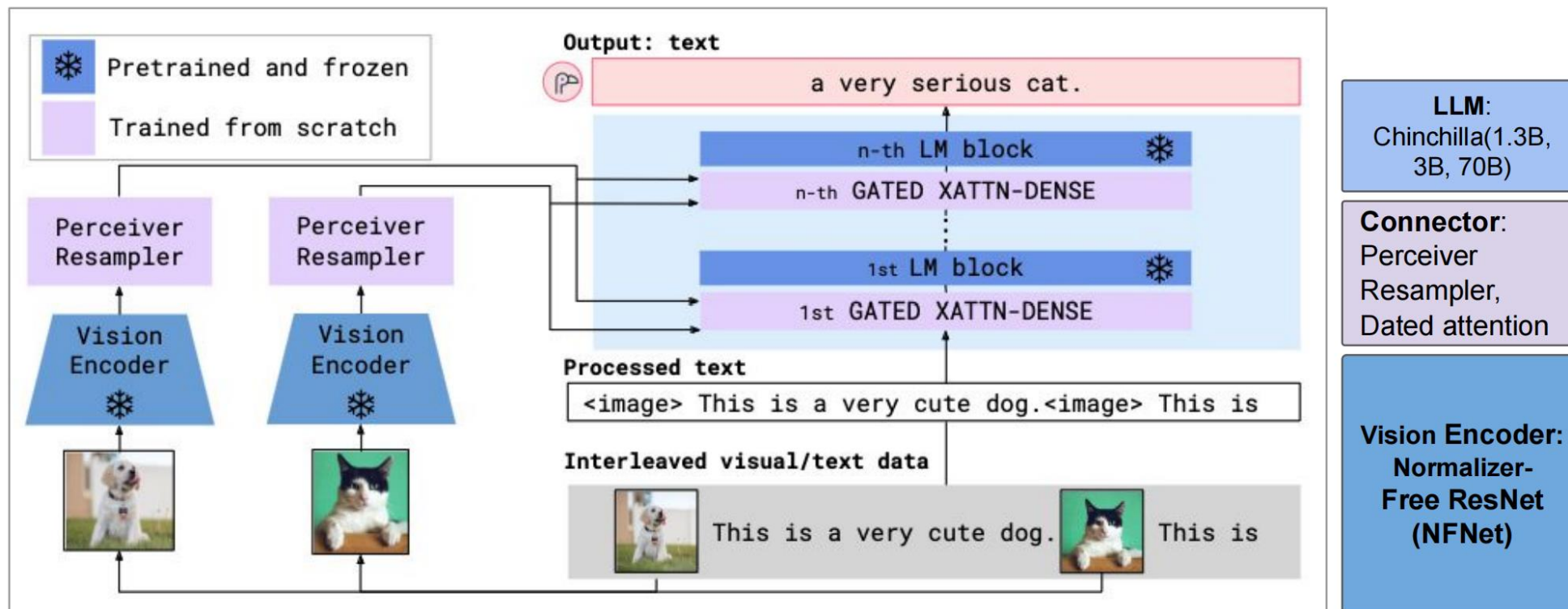
■ CLIP / Flamingo / LLaVA / Others



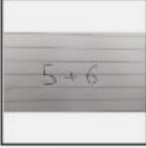
Zero-shot CLIP outperforms linear probe of SimCLR, BiT, ResNet. Probed CLIP also outperforms SOTA computer vision models on both few-shot learning and on 27 image datasets(reported average).

■ CLIP / Flamingo / LLaVA / Others

- Flamingo is **in-context learning visual language model** that marries pretrained vision encoder and language model, can take an arbitrary number of interleaved image(and videos) and texts samples.
- Flamingo achieved strong zero-shot performance on many multimodal downstream tasks such as visual question answering, image captioning, among others.



■ CLIP / Flamingo / LLaVA / Others

Input Prompt					Completion
	This is a chinchilla. They are mainly found in Chile.		This is a shiba. They are very popular in Japan.	 This is	a flamingo. They are found in the Caribbean and South America.
	What is the title of this painting? Answer: The Hallucinogenic Toreador.		Where is this painting displayed? Answer: Louvres Museum, Paris.	 What is the name of the city where this was painted? Answer:	Arles.
	Output: "Underground"		Output: "Congress"	 Output:	"Soulomes"
	2+1=3		5+6=11		3x6=18
	Output: A propaganda poster depicting a cat dressed as French emperor Napoleon holding a piece of...		Output: A pink room with a flamingo pool float.	 Output:	A portrait of Salvador Dali with a robot head.

Flamingo is among the models that populated in-context learning/few shot learning in VLMs!

■ CLIP / Flamingo / LLaVA / Others

Two training stages

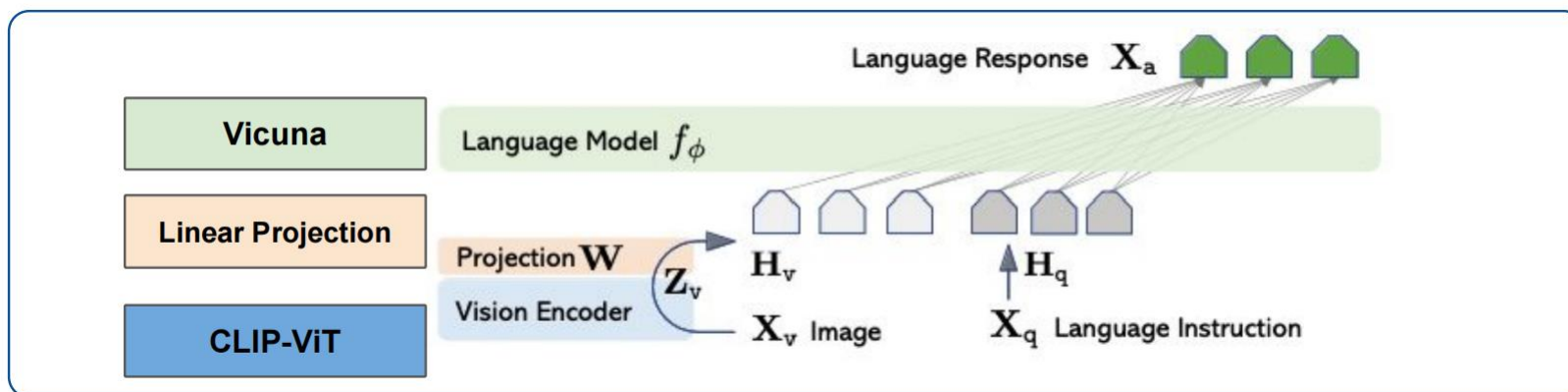
■ Stage 1 - Pre-training for vision and language alignment

Train the model on 595K image-text pairs filtered from CC3M(Common Crawls).
The image-text pairs are converted into instructions.

■ Stage 2 - Supervised fine-tuning

Finetune the base LLaVA on specific tasks: multimodal chats and
multimodal reasoning(with ScienceQA dataset).

In each stage, visual encoder is frozen. Projection layer and LLM are continually updated!



■ CLIP / Flamingo / LLaVA / Others

Visual input example, Extreme Ironing:



Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

User
LLaVA

What is unusual about this image?

The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.

User
LLaVA

[Start a new conversation, and clear the history]

What's happening in the scene?

The image depicts a man standing on top of a yellow SUV in a busy city street. He is holding a portable ladder, seemingly ironing clothes while standing on the vehicle. Around the scene, there are other cars, a traffic light, and a fire hydrant. Additionally, there are two other people in the vicinity, one close to the man on the car and another further away. The man on the ladder appears to be attracting attention due to his unconventional choice of ironing his clothes on top of a moving car. The city street around him is bustling with activity, adding to the unique nature of the scene.

User
GPT-4 [36]

What is unusual about this image?

The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

User
BLIP-2

What is unusual about this image?

a man is sitting on the back of a yellow cab

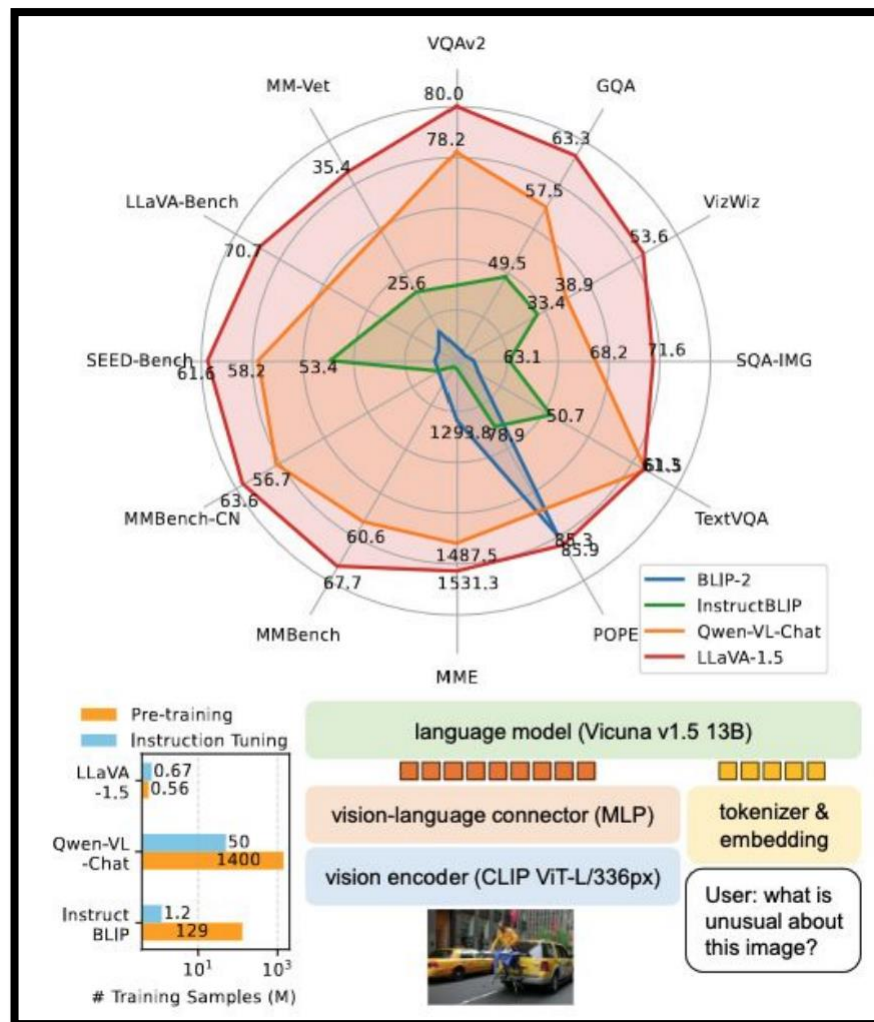
User
OpenFlamingo

What is unusual about this image?

The man is drying his clothes on the hood of his car.

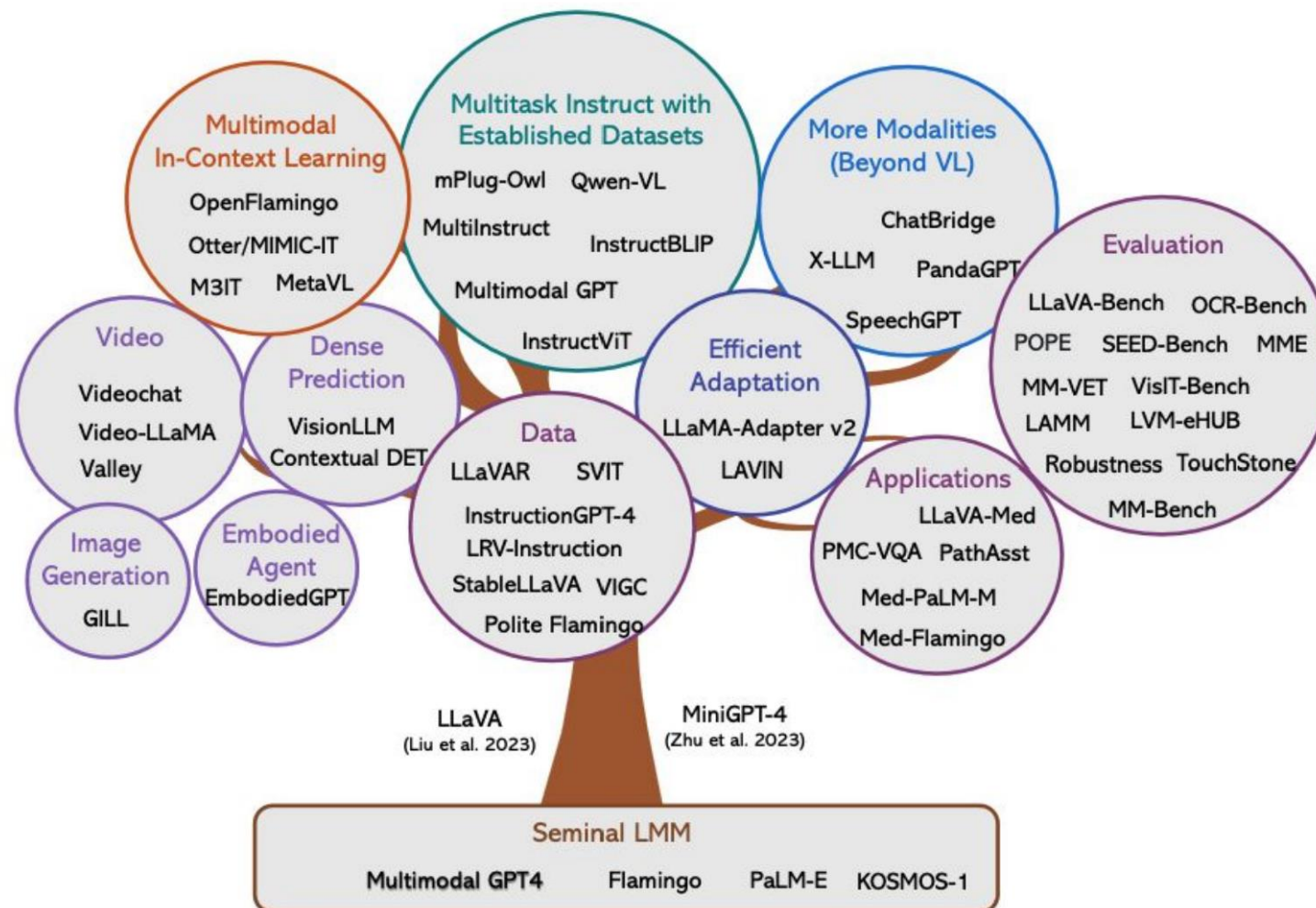
LLaVA generates reasonable responses when tasked with complex reasoning problems!

■ CLIP / Flamingo / LLaVA / Others



LLaVA has improved significantly since the first release.

■ CLIP / Flamingo / LLaVA / Others

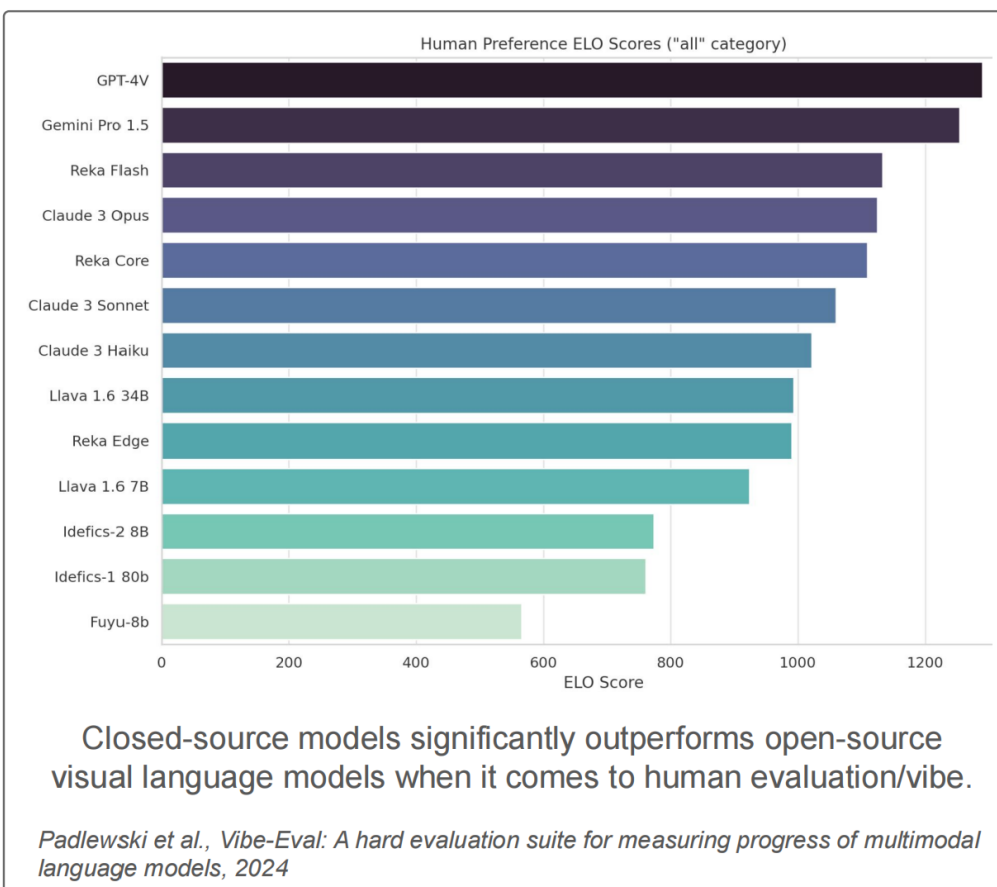
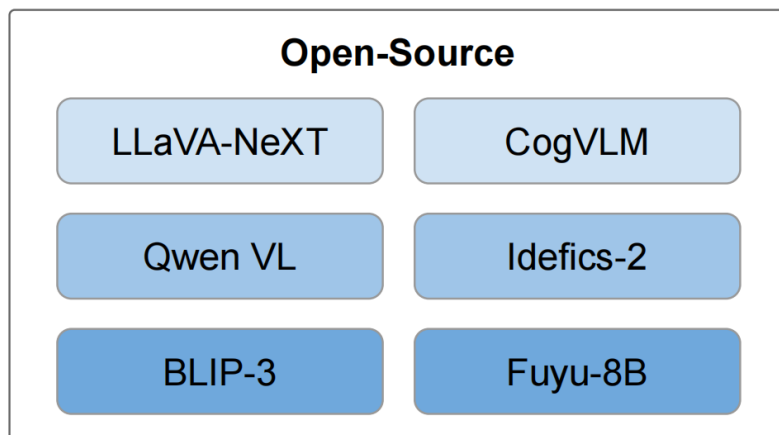
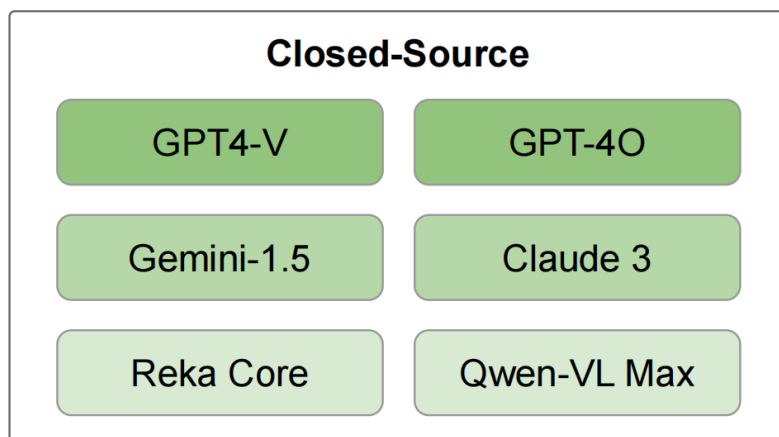


■ CLIP / Flamingo / LLaVA / Others

Open-source



Closed Multimodal LLMs



Challenges

Models and benchmarks often exhibit language biases, enabling LLMs to solve them without image analysis.

The current approach of combining vision encoders with LLMs using language modeling objectives for training MLLMs may not be optimal, indicating room for improvement.

New Directions

- Developing multimodal agents and evaluation benchmarks.
- Pursuing vertical integration, such as in healthcare.
- Creating efficient models that perform better on real-world tasks.
- Scaling more modalities with a unified architecture across vision, sound, etc.

Bridging Perception   and Action  :

Vision-Language-Action (VLA) Models



1

VLA Preliminary

2

VLA Model

- Implicit End-to-End VLA
- Explicit End-to-End VLA
- Hierarchical End-to-End VLA
- Specialized End-to-End VLA

3

VLA Discussion

1

VLA Preliminary

- VLA Overview
- Dataset & Metric
- Simulation Environment

■ VLA Overview / Simulation Environment / Dataset & Metric

Action



Traditional Robot **Decision**



Vision + Language



Traditional **Perception**



Vision + Language + Action



Perception-Decision Integrated Robot

■ VLA Overview / Simulation Environment / Dataset & Metric

Industrial Progress



Galbot



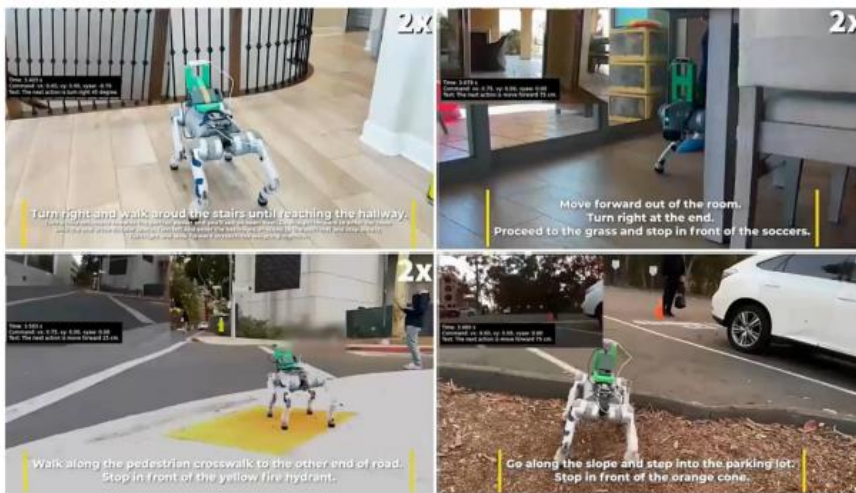
Unitree



Fourier

■ VLA Overview / Simulation Environment / Dataset & Metric

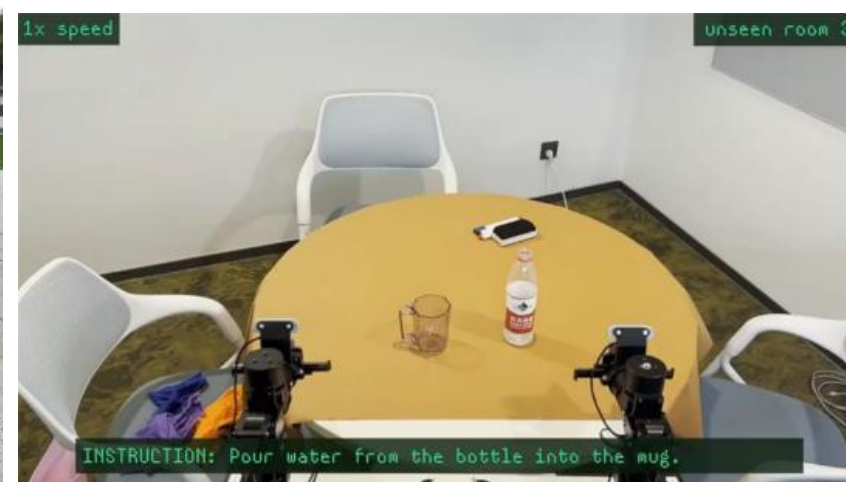
Academic Progress



Quadruped robot



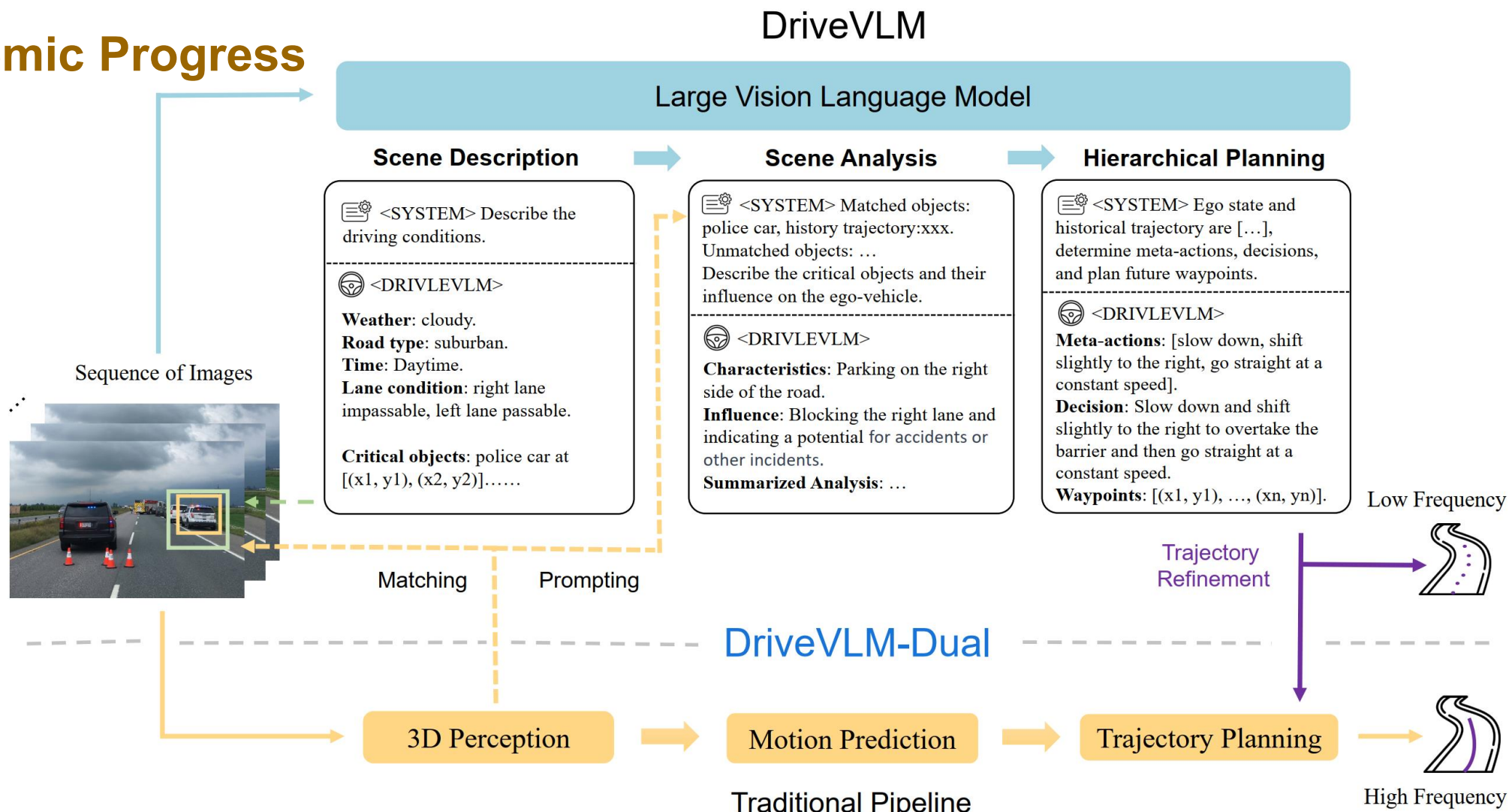
Humanoid



Robotic Arm

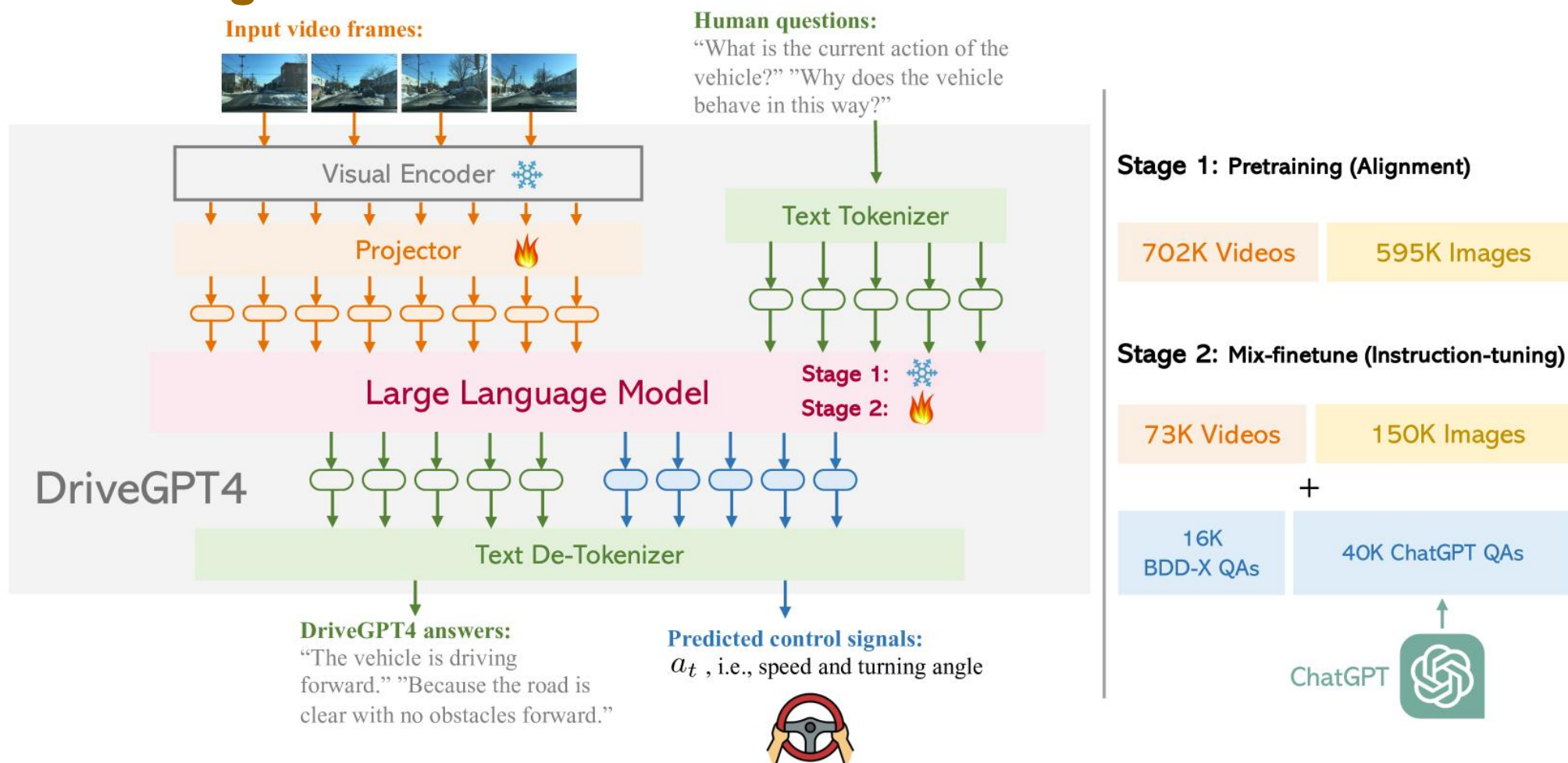
■ VLA Overview / Simulation Environment / Dataset & Metric

Academic Progress



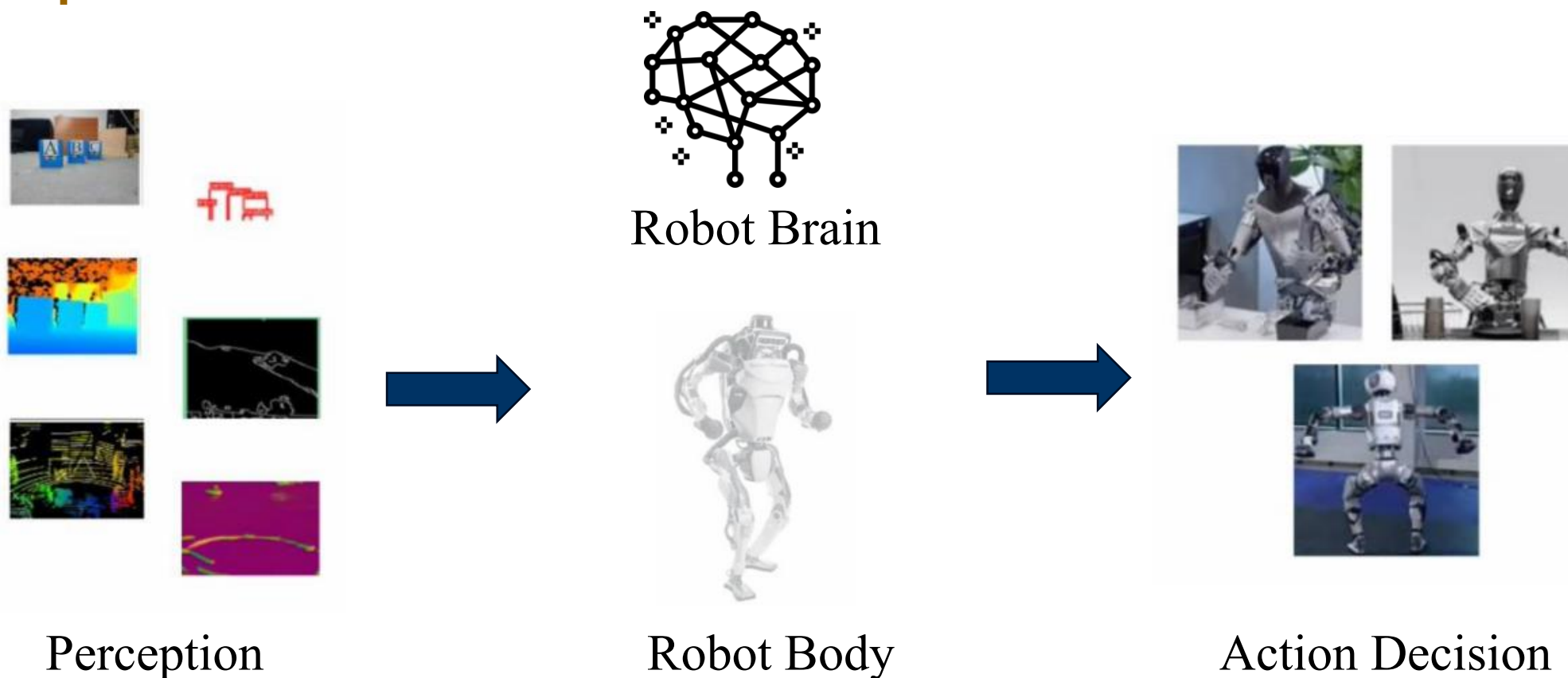
■ VLA Overview / Simulation Environment / Dataset & Metric

Academic Progress



■ VLA Overview / Simulation Environment / Dataset & Metric

Overall process



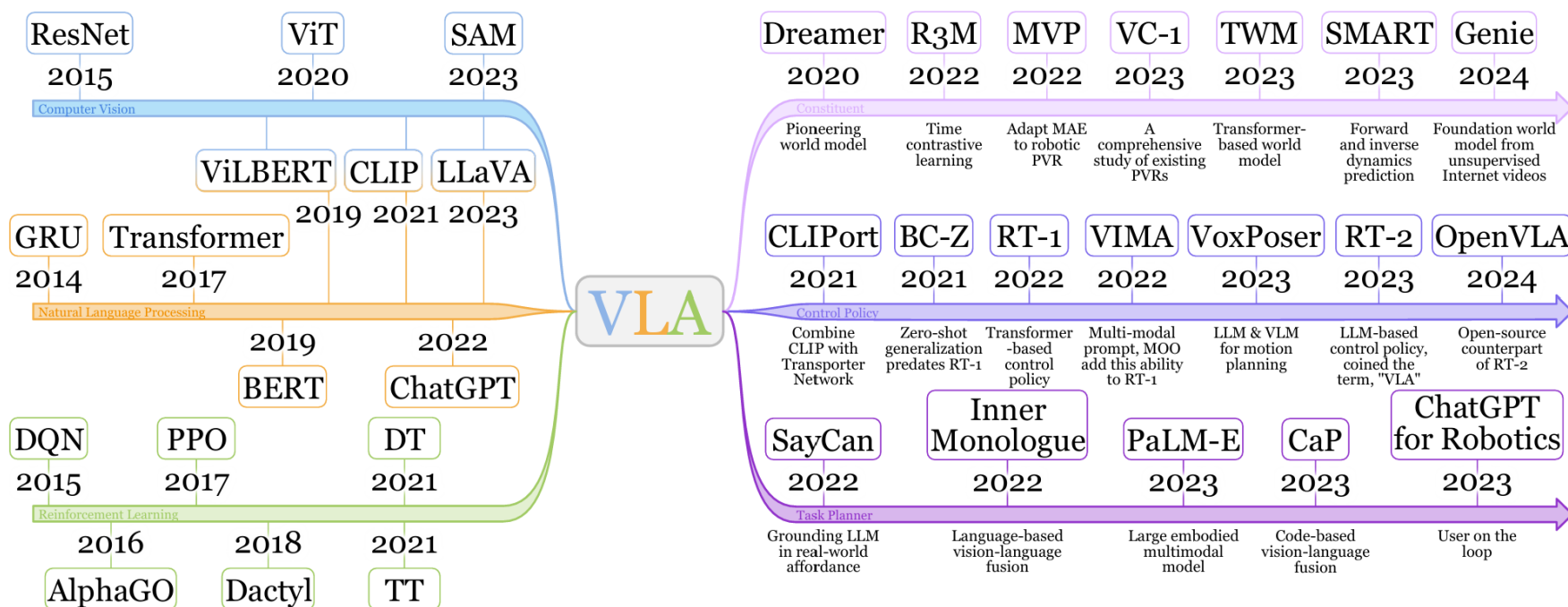
■ VLA Overview / Simulation Environment / Dataset & Metric

End-to-end Model driving



■ VLA Overview / Simulation Environment / Dataset & Metric

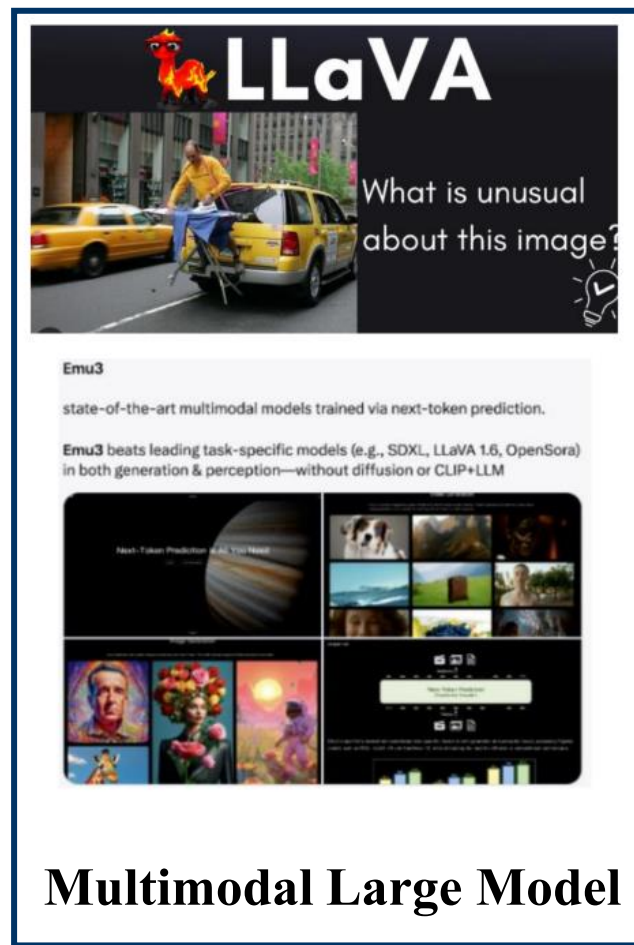
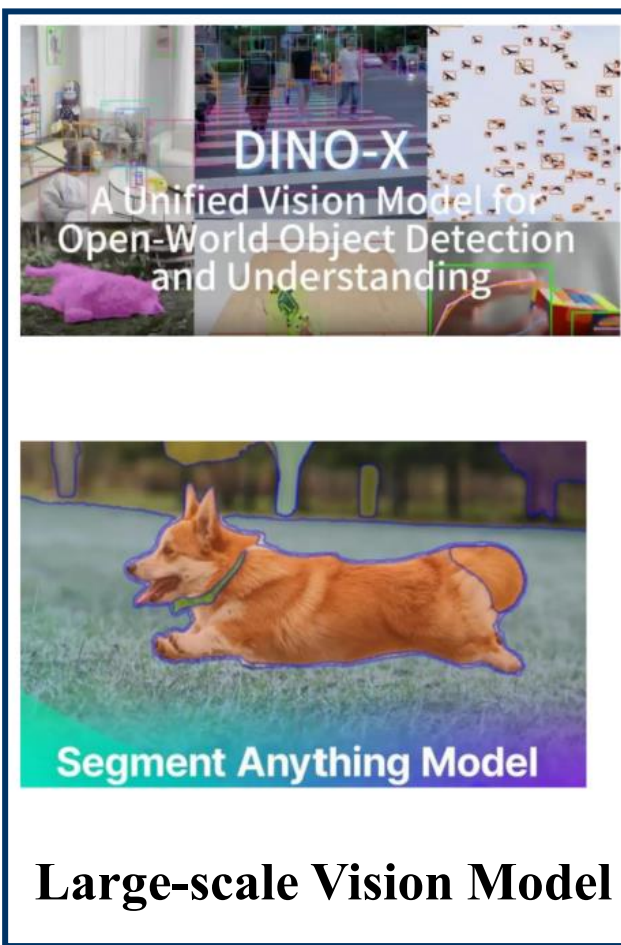
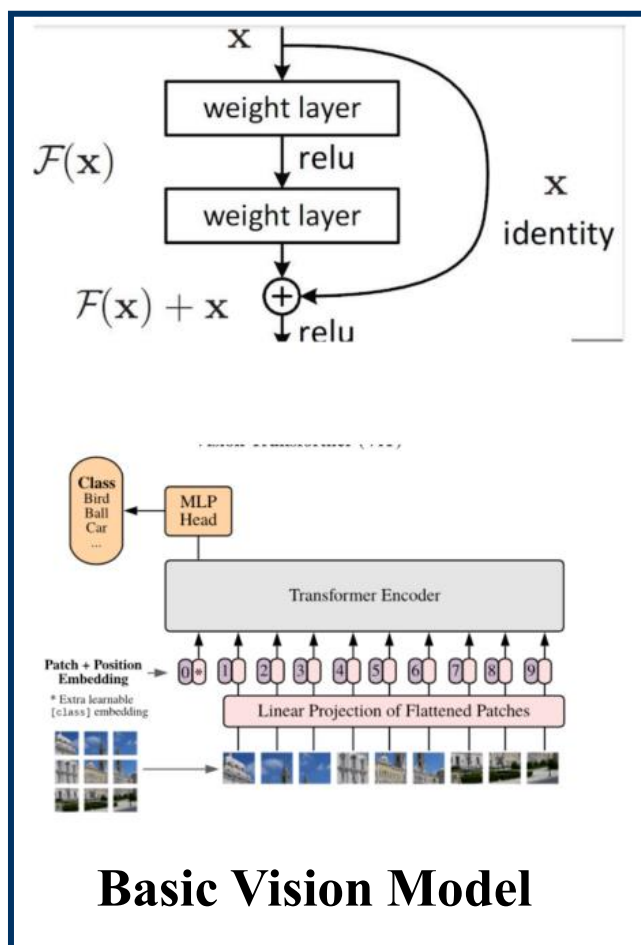
Definition



VLA is a new paradigm in the field of embodied intelligence, which directly generates robot-executable actions from given **language instructions** and **visual signals**.

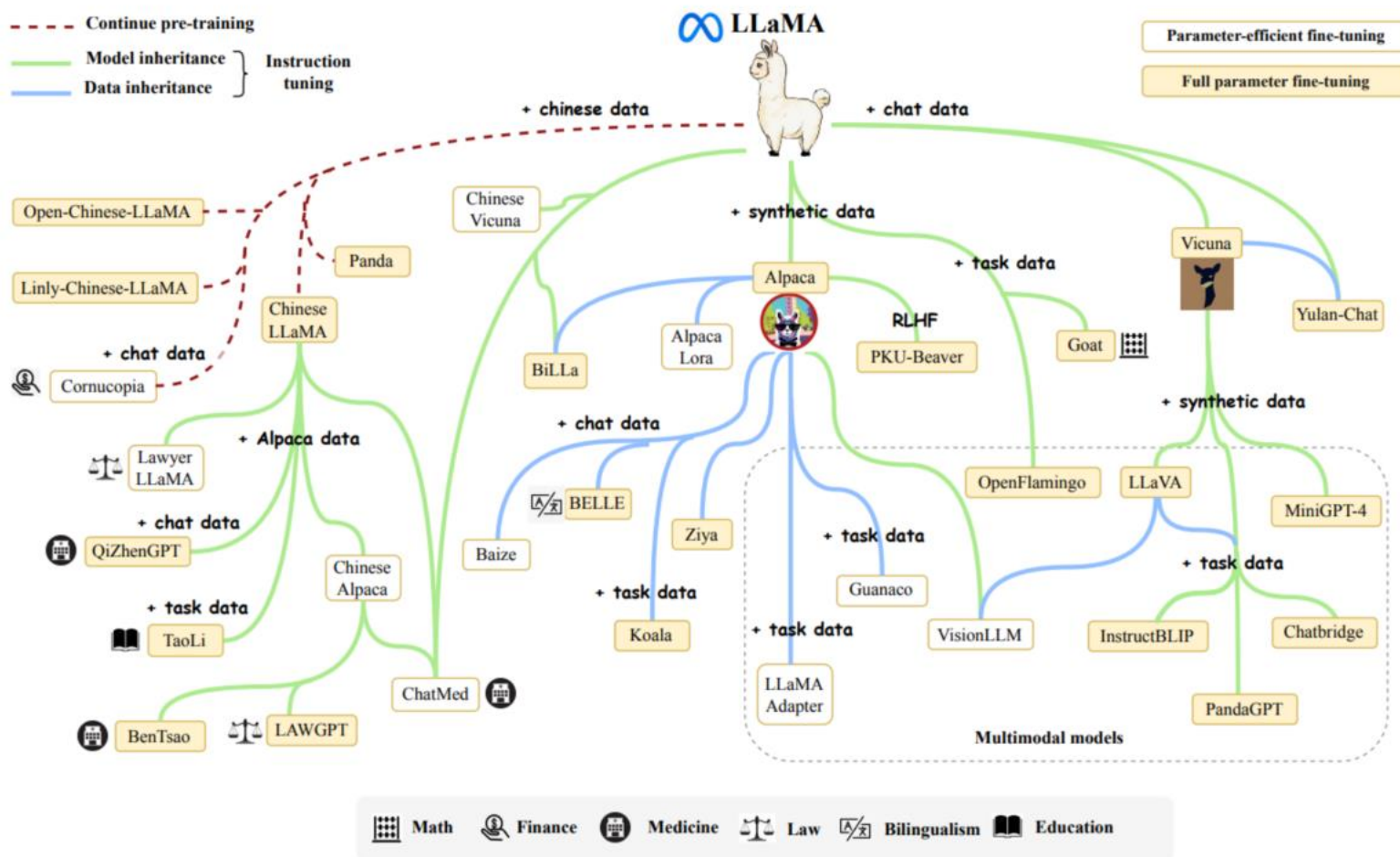
■ VLA Overview / Simulation Environment / Dataset & Metric

VLA Elements — Vision



■ VLA Overview / Simulation Environment / Dataset & Metric

VLA Elements — Language

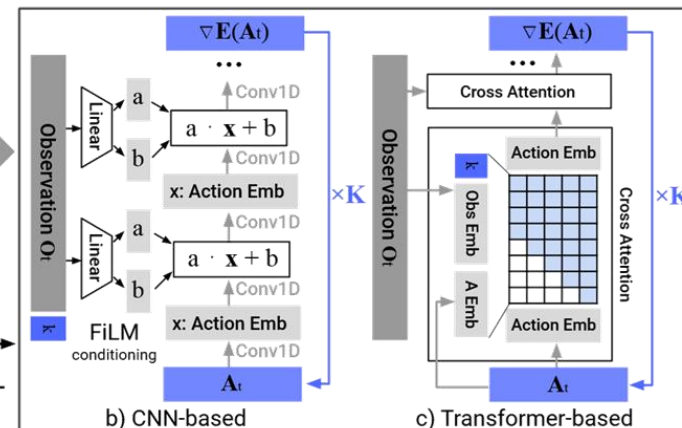
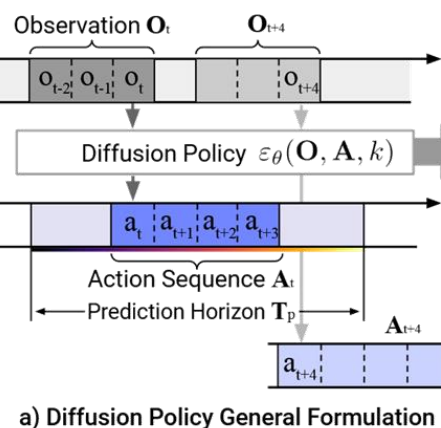
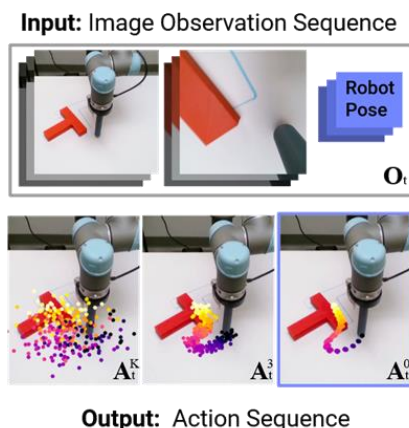


■ VLA Overview / Simulation Environment / Dataset & Metric

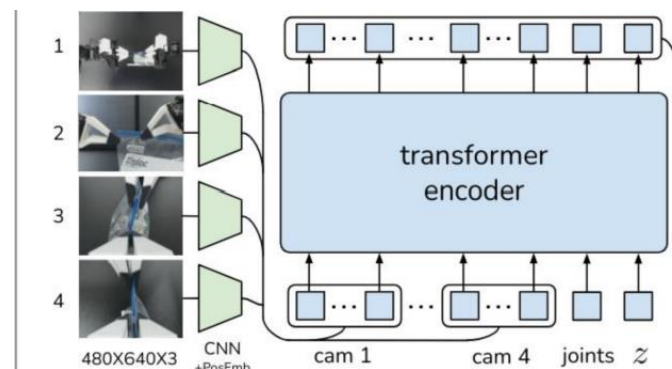
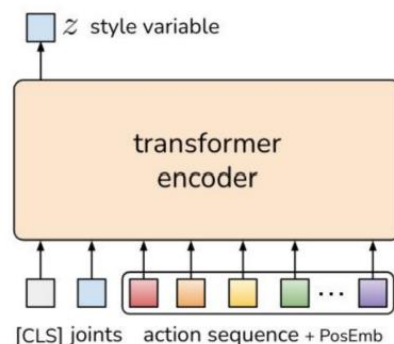
VLA Elements — Action

Vision-Action Model

Diffusion Policy

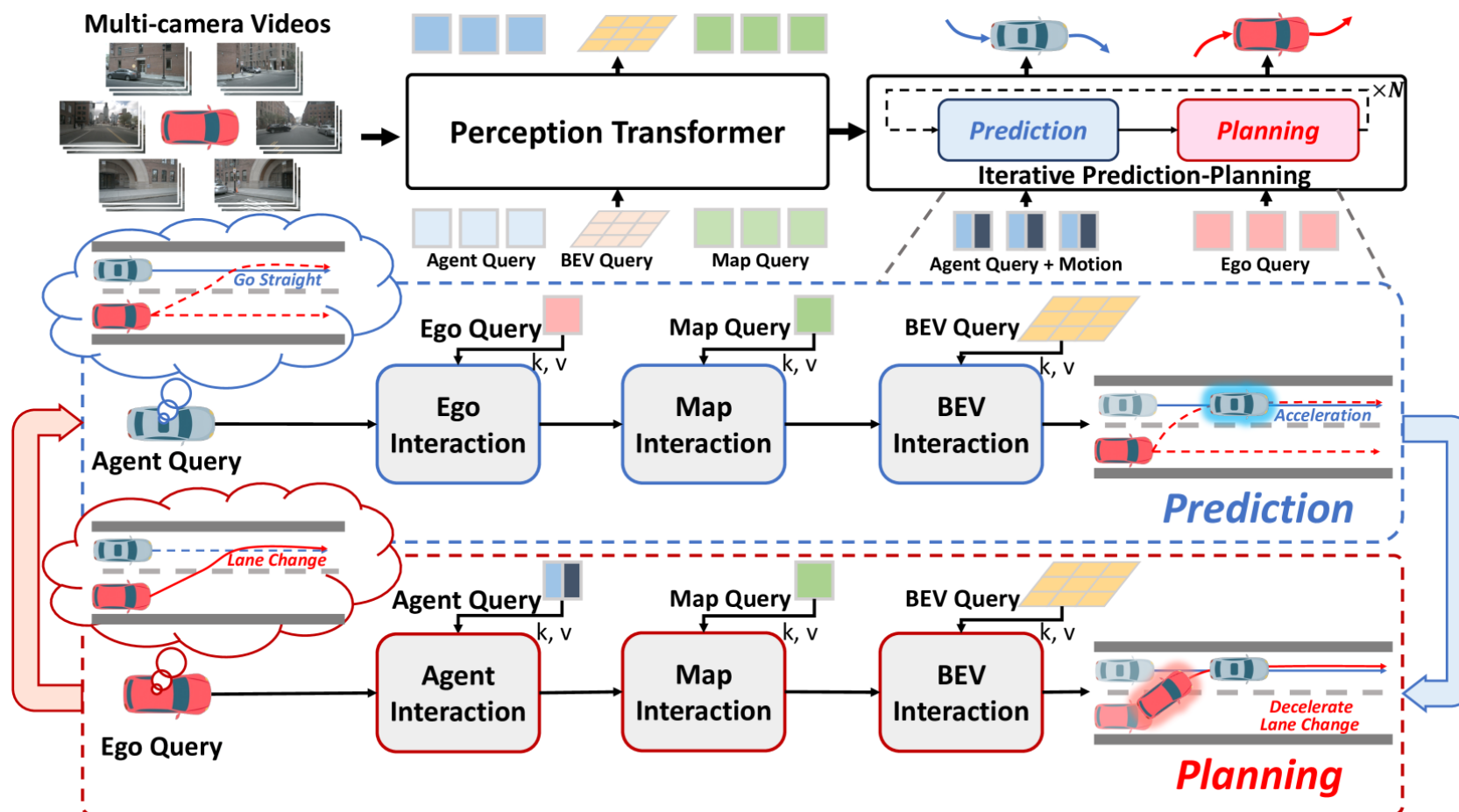


Action Chunking with Transformers (ACT)



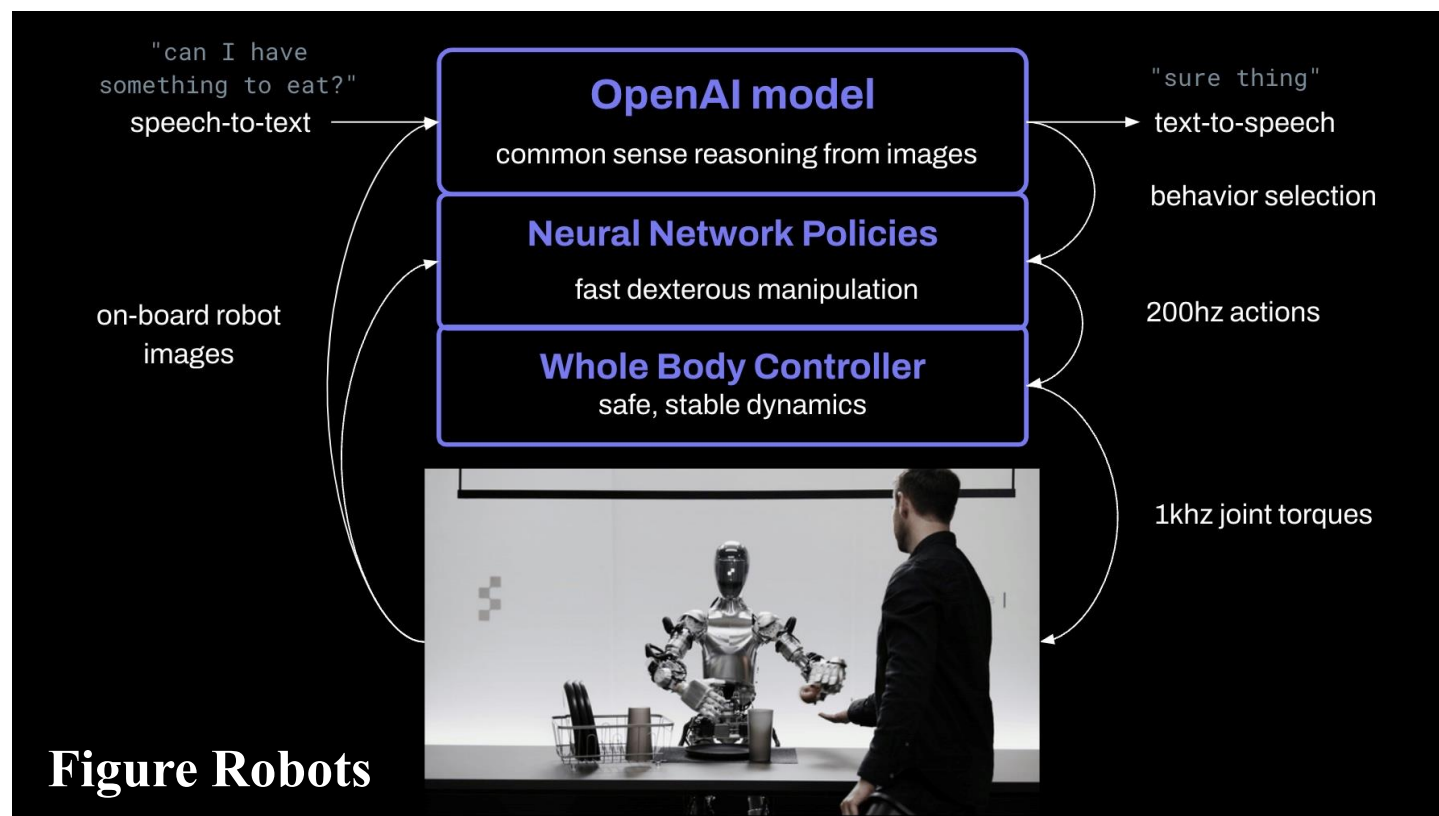
■ VLA Overview / Simulation Environment / Dataset & Metric

VLA Elements — End-to-end training



■ VLA Overview / Simulation Environment / Dataset & Metric

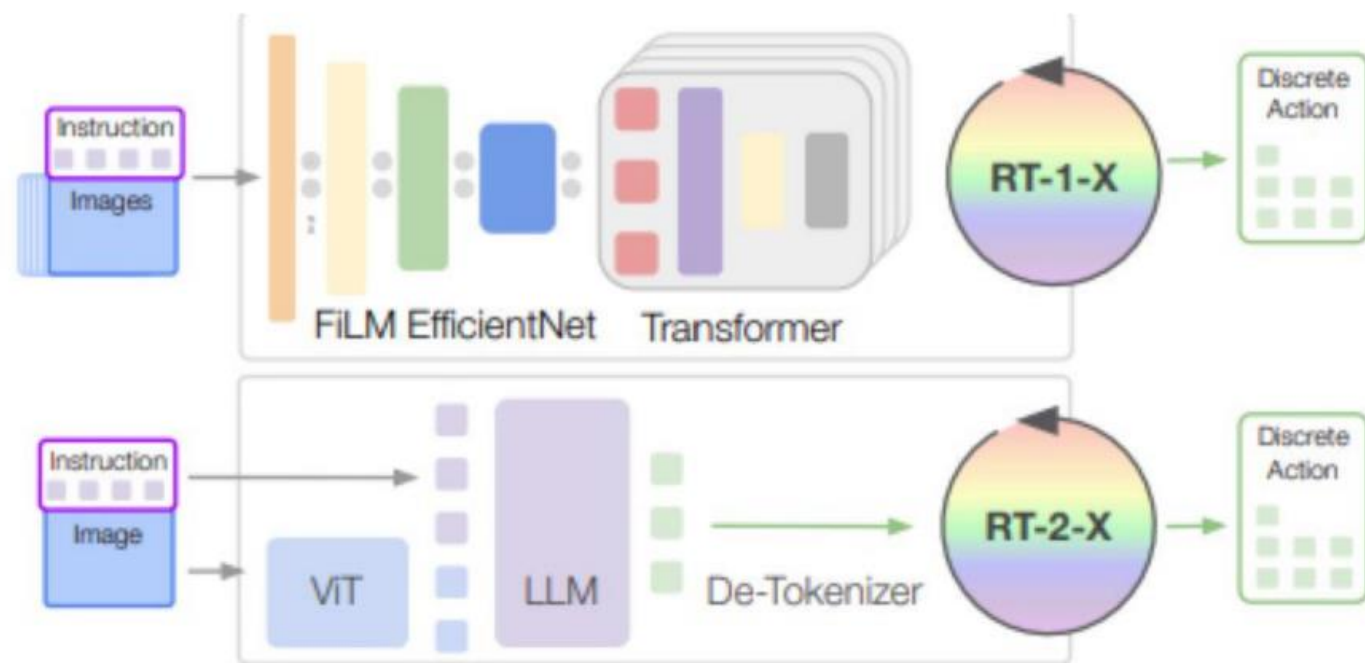
VLA Solution — Layered Approach used in Industry



- Perception and decision-making are **trained separately** and **used in layers**.
- Perception **doesn't enhance decision-making** during training.
- Final decisions are **limited to existing actions** in the skill library.

■ VLA Overview / Simulation Environment / Dataset & Metric

VLA Solution — End-to-End Approach used in Academy

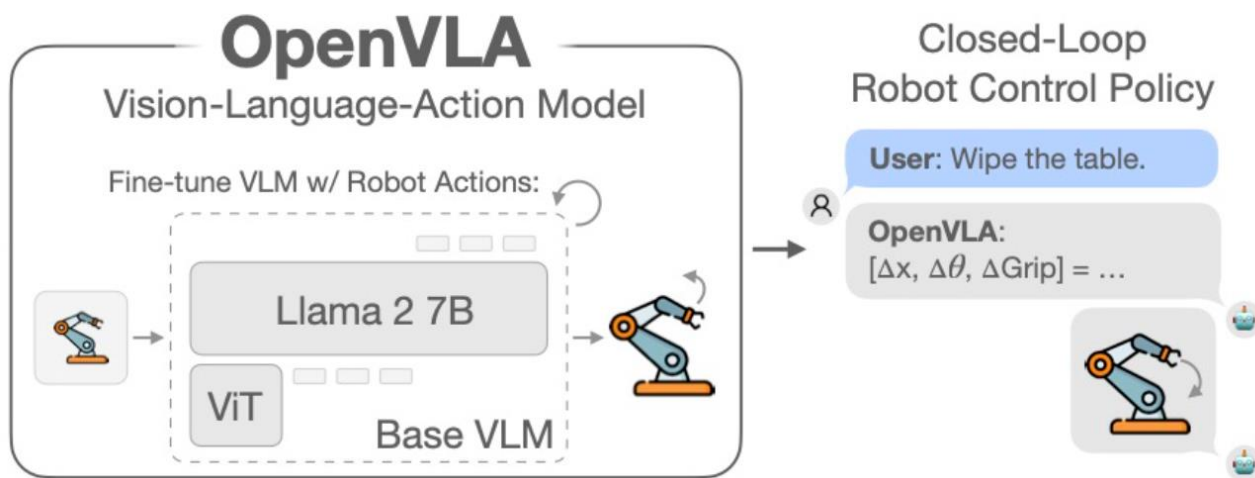


DeepMind

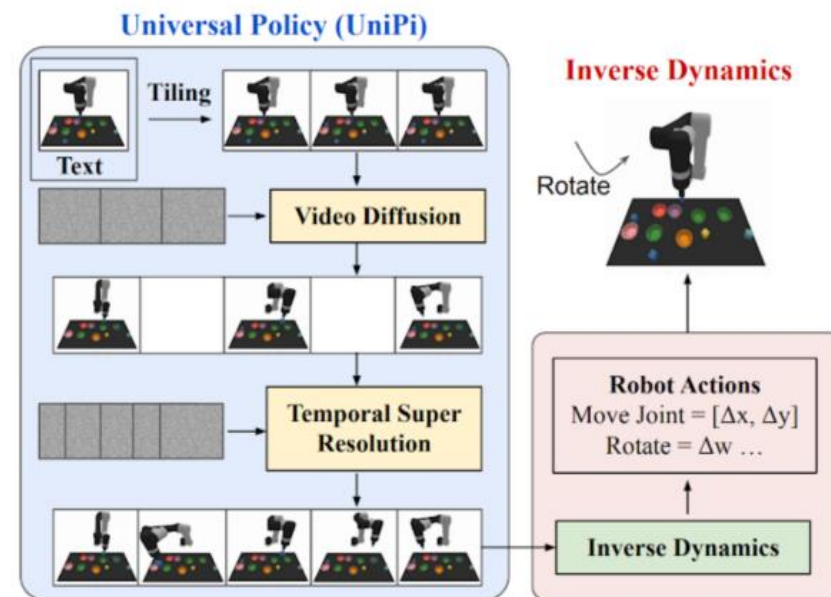
- Perception and decision-making are **trained end-to-end**, generating actions in one step.
- Leverage the strong research foundations of Vision and Language to enhance the generation of Action.

■ VLA Overview / Simulation Environment / Dataset & Metric

VLA Solution — End-to-End Approach used in Academy



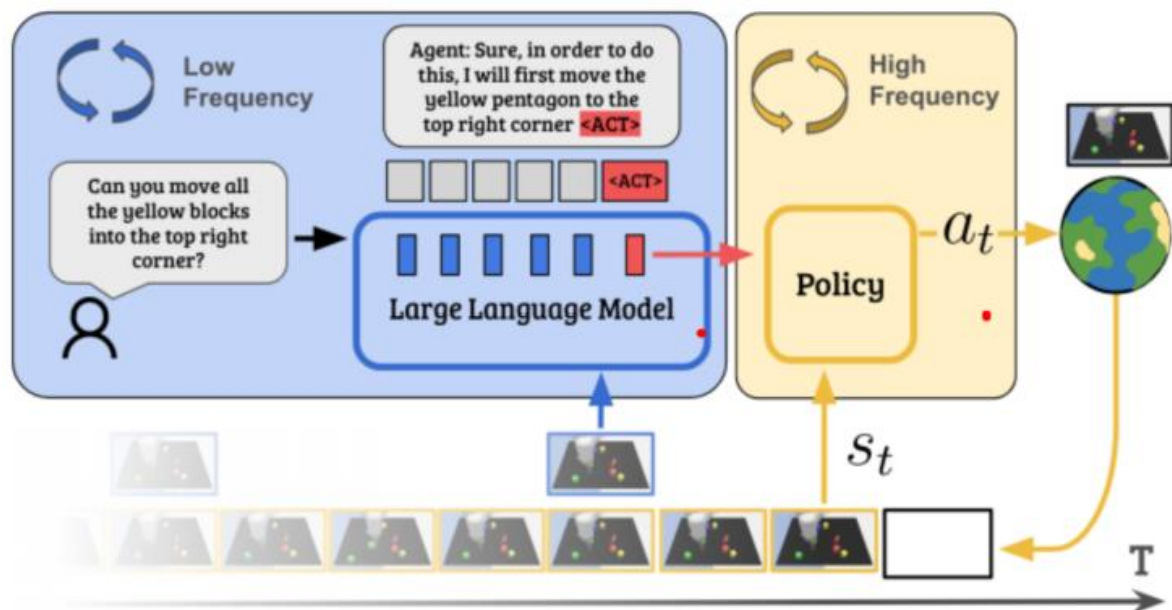
Implicit end-to-end VLA



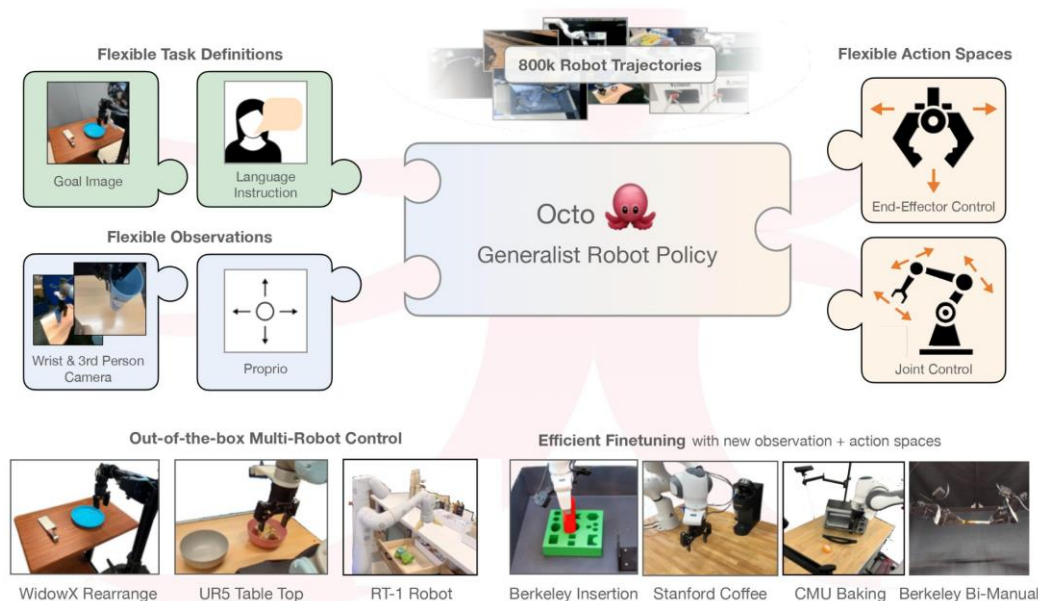
Explicit end-to-end VLA

■ VLA Overview / Simulation Environment / Dataset & Metric

VLA Solution — End-to-End Approach used in Academy



Hierarchical end-to-end VLA



Others

■ VLA Overview / Simulation Environment / Dataset & Metric

Applicable Configurations

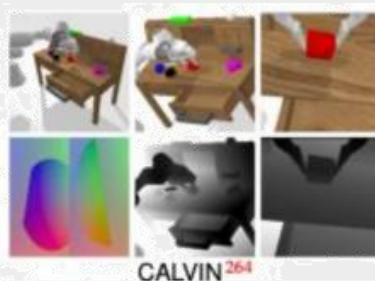


RoboCasa



GR-Scenes

Full Configuration



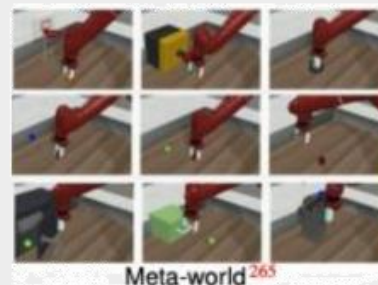
CALVIN²⁶⁴



VIMAbench²⁴⁷



Robotic Arm



Meta-world²⁶⁵



habitat

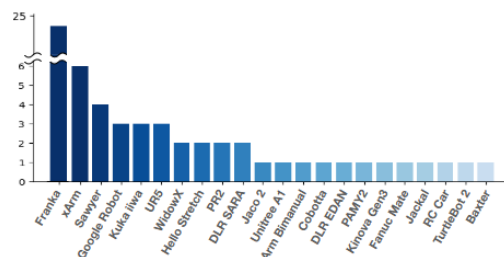
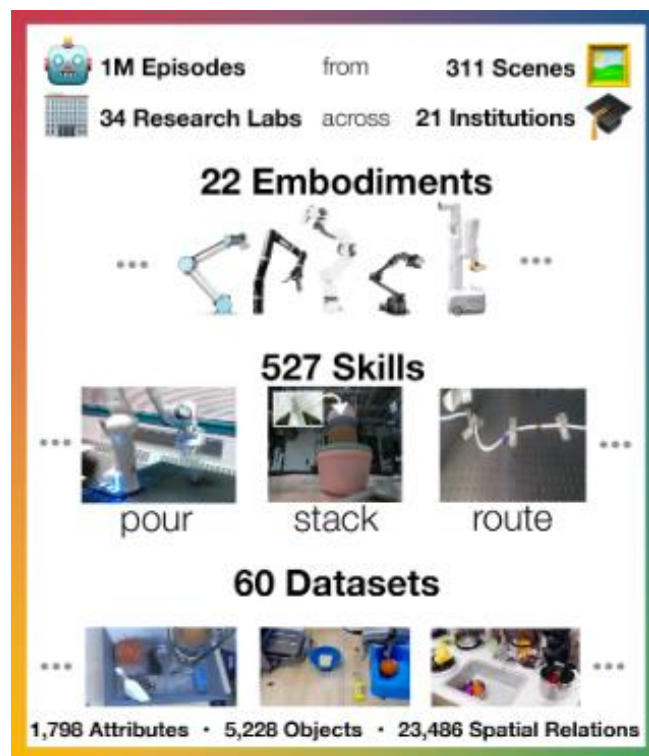


isacgym

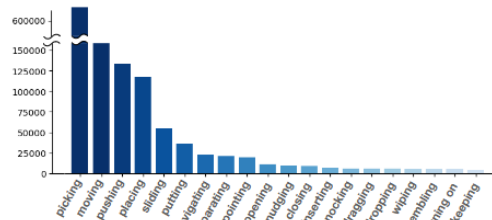
Wheeled / Legged

■ VLA Overview / Simulation Environment / Dataset & Metric

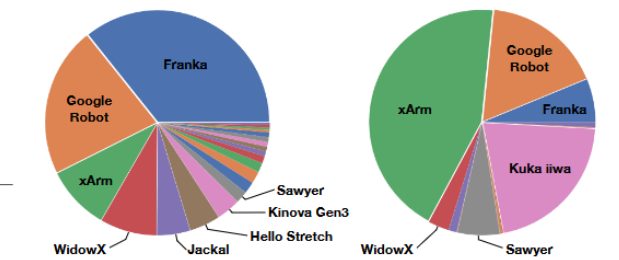
Dataset — Open X-Embodiment



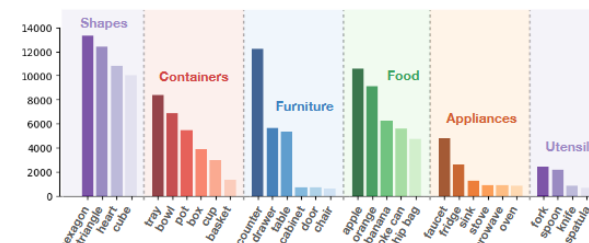
(a) # Datasets per Robot Embodiment



(d) Common Dataset Skills



(b) # Scenes per Embodiment (c) # Trajectories per Embodiment



(e) Common Dataset Objects

1M+ real robot trajectories spanning 22 robot embodiments

■ VLA Overview / Simulation Environment / Dataset & Metric

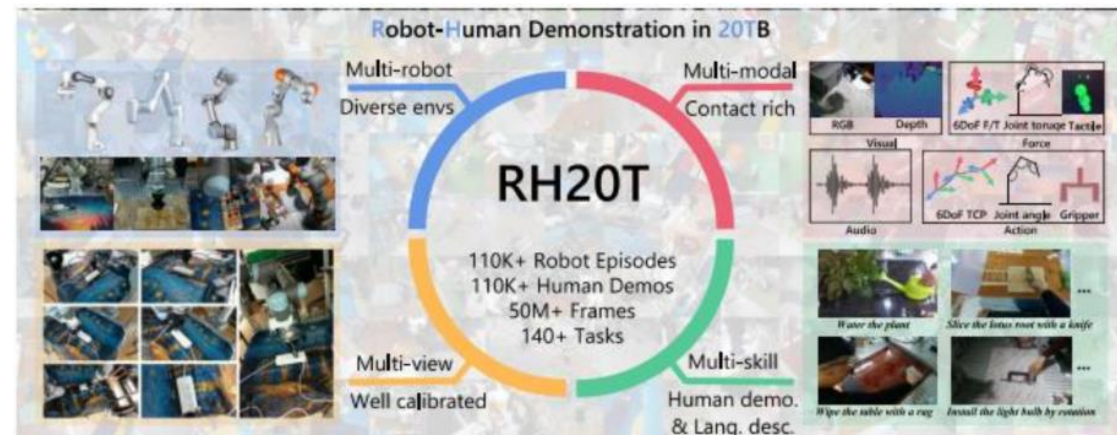
Dataset — RH-20T

Task. Insert the plug into the socket.



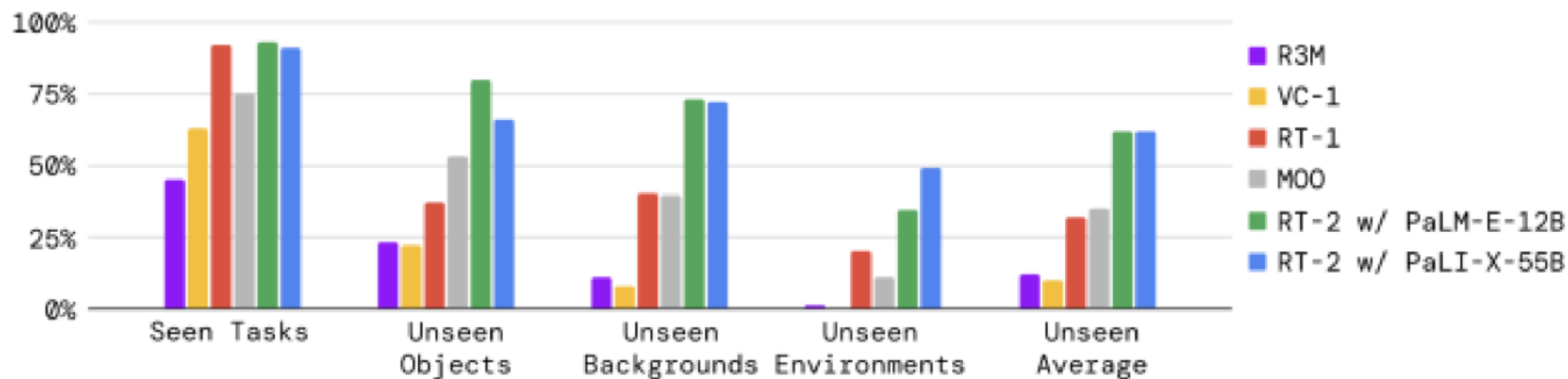
Human Demonstration
(multiple views, only one is demonstrated here)

Robot Manipulation (multiple views)



■ VLA Overview / Simulation Environment / Dataset & Metric

Metric — Success Rate in Different Dimensions



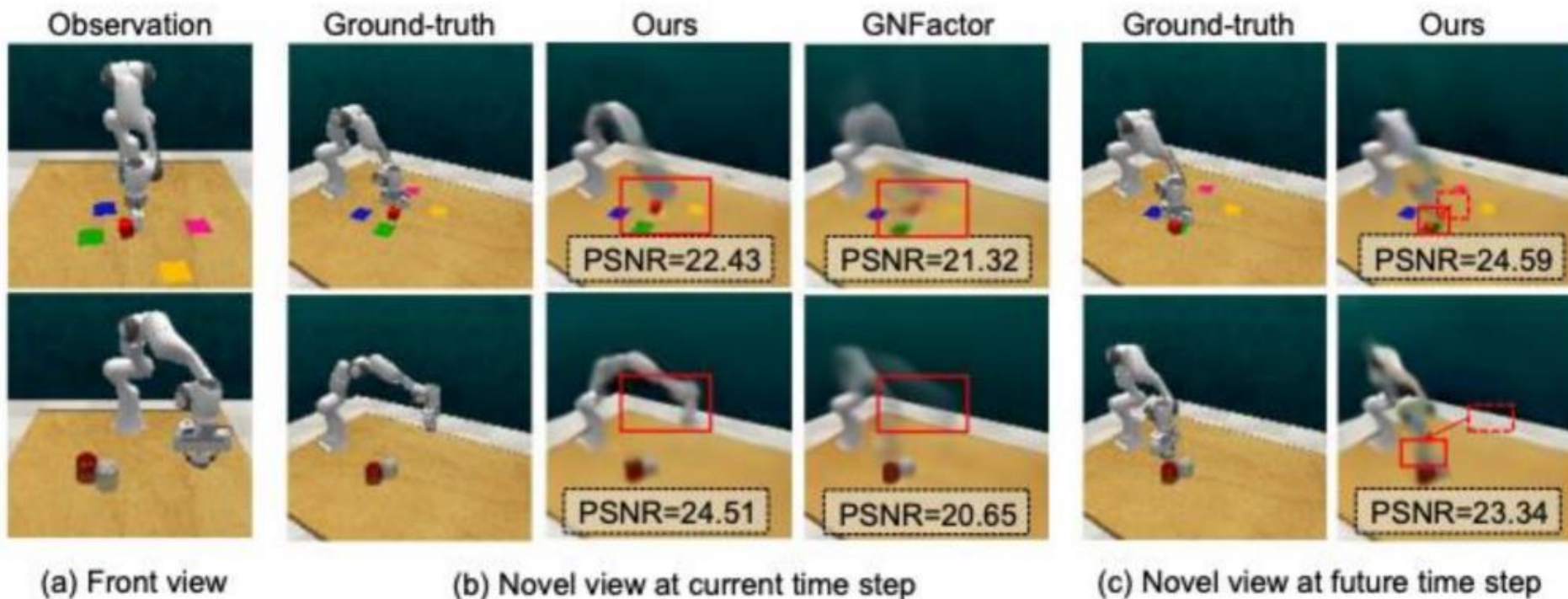
■ VLA Overview / Simulation Environment / Dataset & Metric

Metric — Average Length

Method	Input	MTLC	LH-MTLC						
		(32 tasks)	No. Instructions in a Row (1000 chains)						
			1	2	3	4	5	Avg. Len.	
<u>RoboUniView</u>	Static RGB-D + Gripper RGB-D + Cam params	-	96.2%	88.8%	77.6%	66.6%	56.3%	3.85	
<u>MDT</u>	Static RGB + Gripper RGB	-	93.7%	84.5%	74.1%	64.4%	55.6%	3.72	
<u>HULC++</u>	Static RGB + Gripper RGB	-	93%	79%	64%	52%	40%	3.3	
<u>LCD</u>	Static RGB + Gripper RGB	74.01%	88.7%	69.9%	54.5%	42.7%	32.2%	2.88	
<u>DeeR</u>	Static RGB + Gripper RGB	-	85.3%	69.6%	54.9%	42.0%	31.2%	2.83	
<u>SPIL</u>	Static RGB + Gripper RGB	-	84.6%	65.1%	50.8%	38.0%	28.6%	2.67	
<u>HULC</u>	Static RGB + Gripper RGB	-	82.7%	64.9%	50.4%	38.5%	28.3%	2.64	
Baseline + delta actions	Static RGB	-	76.4%	48.8%	30.1%	18.1%	9.3%	1.82	
Baseline	Static RGB	53.9%	48.9%	12.9%	2.6%	0.5%	0.08%	0.64	
Baseline	Static RGB + Gripper RGB	51.8%	34.4%	5.8%	1.1%	0.2%	0.08%	0.41	

■ VLA Overview / Simulation Environment / Dataset & Metric

Metric — PSNR / FVD



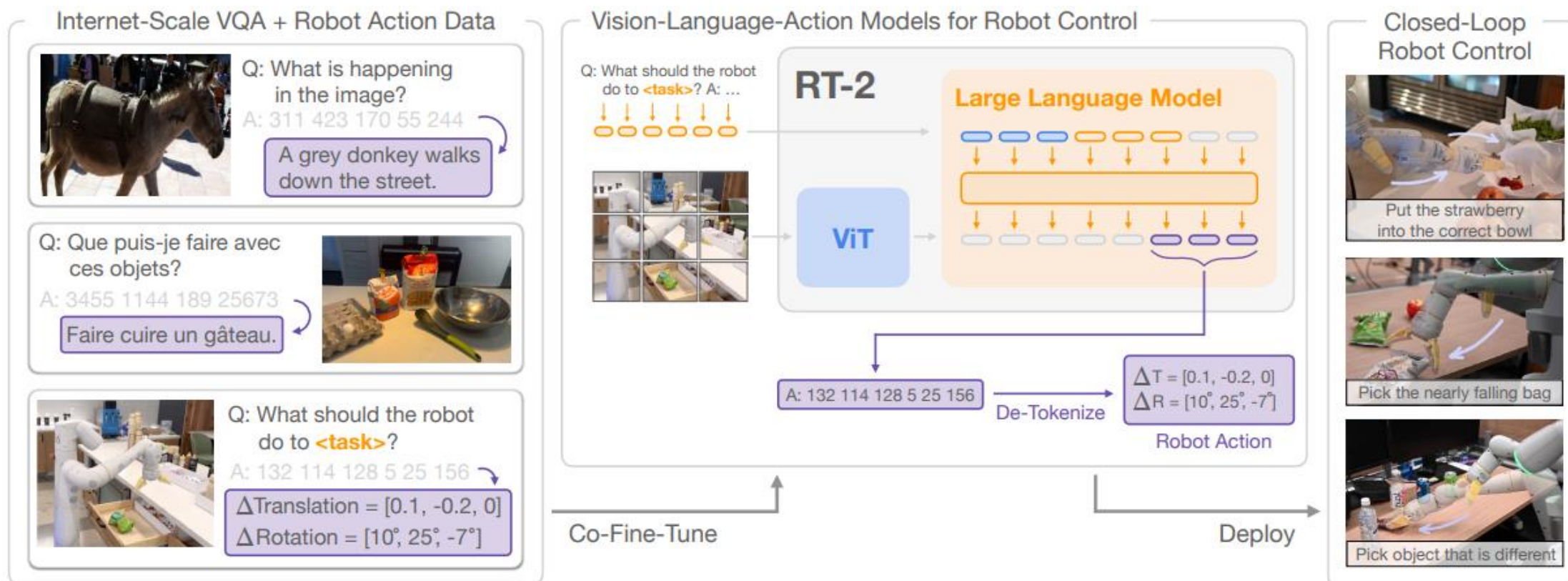
2

VLA Model

- **Implicit End-to-End VLA**
- **Explicit End-to-End VLA**
- **Hierarchical End-to-End VLA**
- **Specialized End-to-End VLA**

■ Implicit / Explicit / Hierarchical / Specialized End-to-End VLA

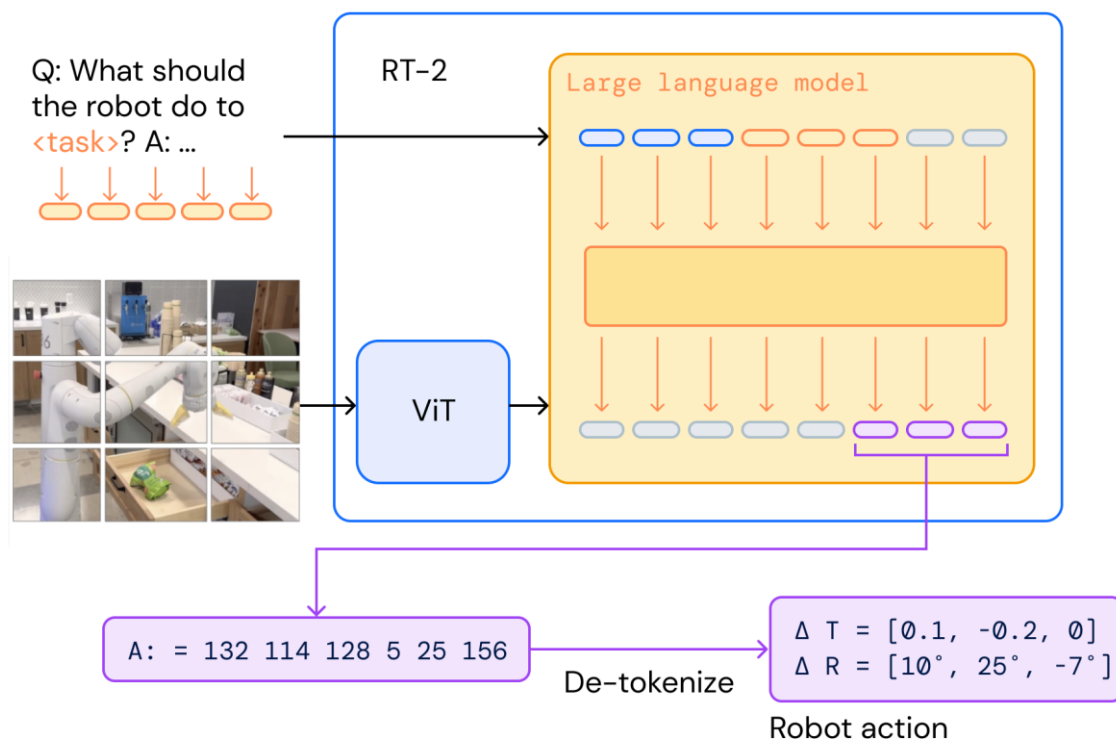
RT-2



■ Implicit / Explicit / Hierarchical / Specialized End-to-End VLA

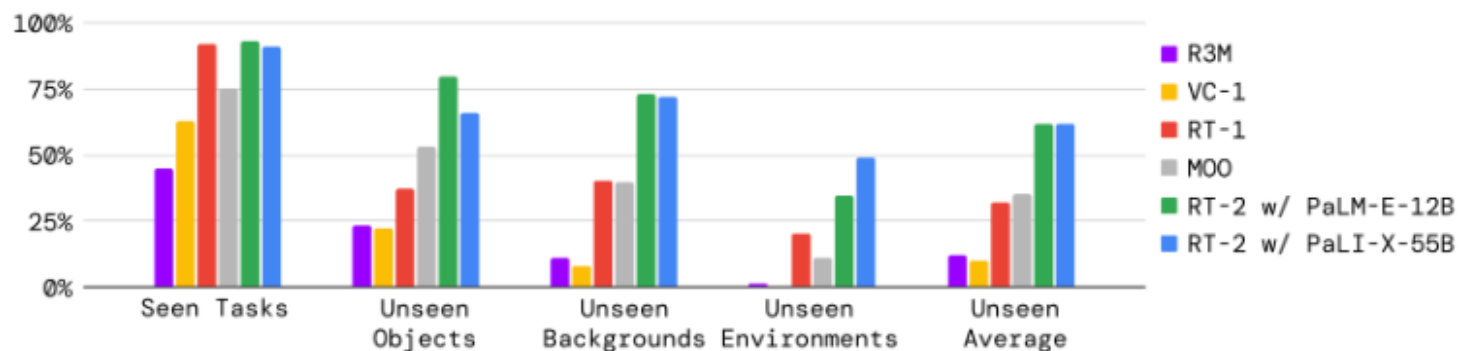
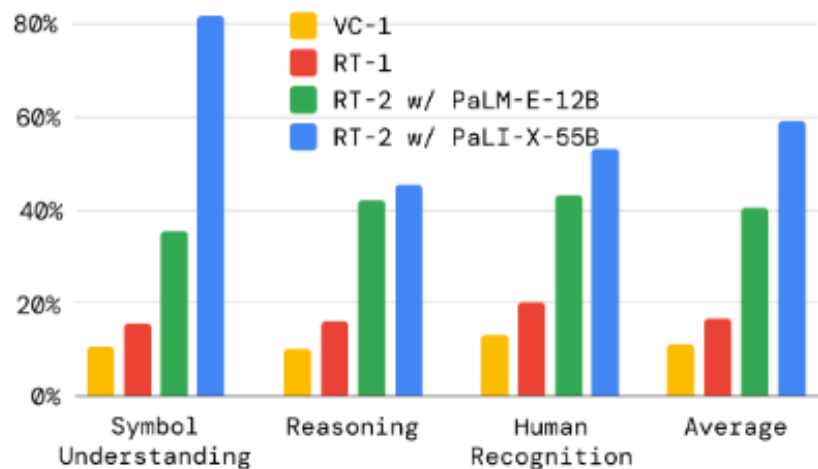
RT-2 — Framework

Represent actions as **text strings**; represent the action prediction as **language translation**.



■ Implicit / Explicit / Hierarchical / Specialized End-to-End VLA

RT-2 — Results

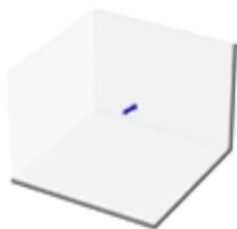


■ Implicit / Explicit / Hierarchical / Specialized End-to-End VLA

RT-2 — Results

move vw to germany

move corvette to the US

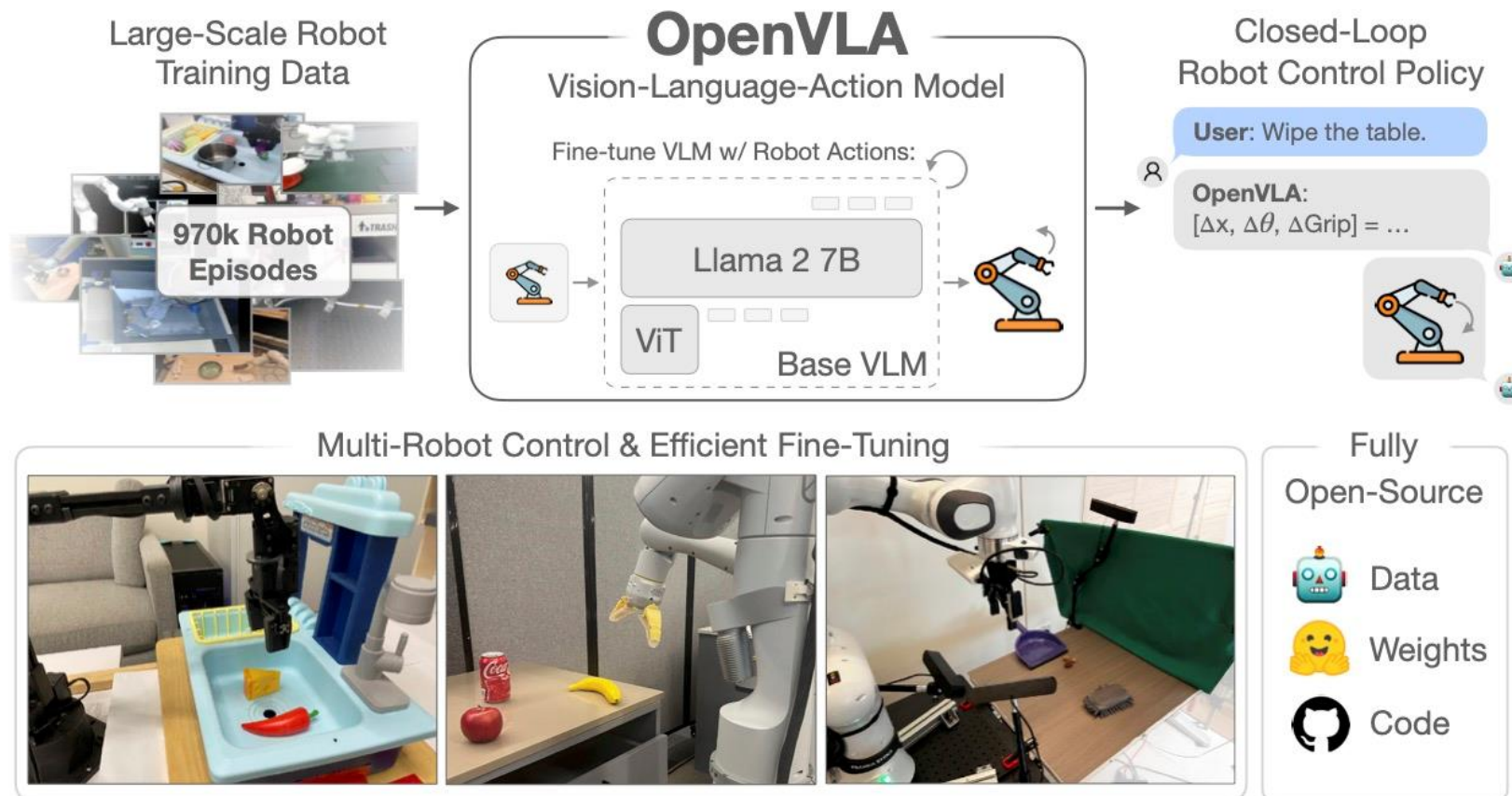


move vw to germany

delta

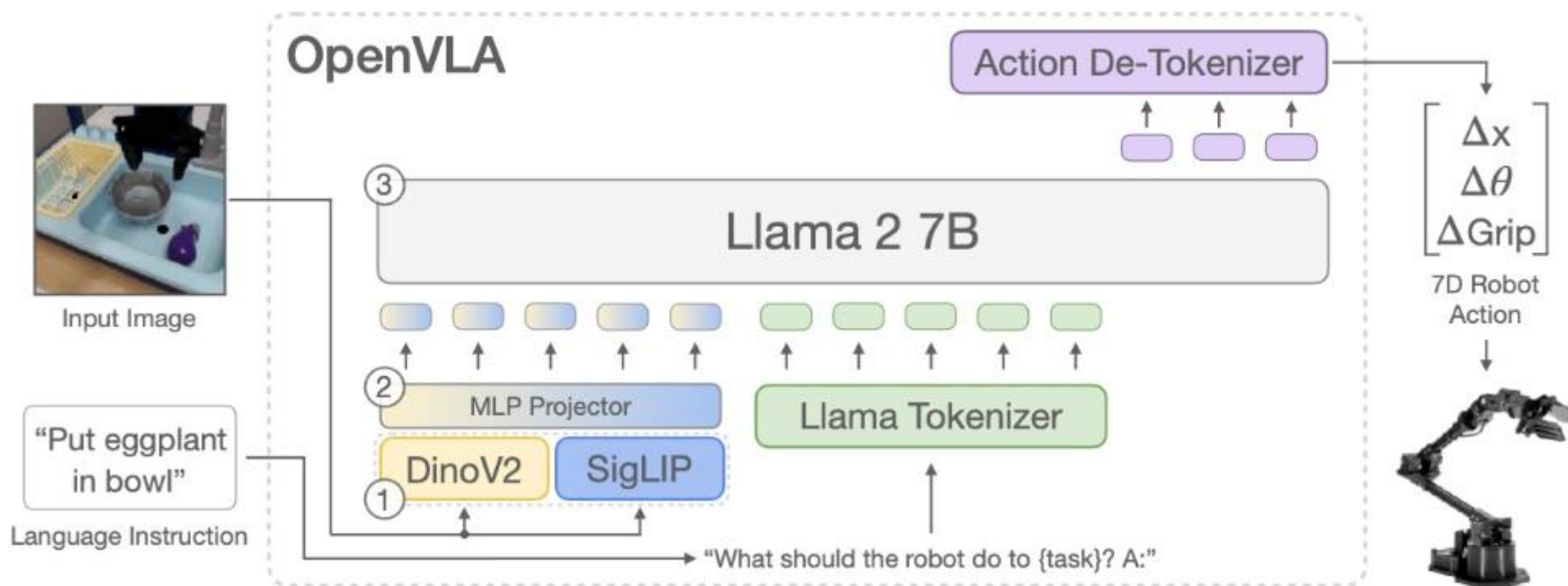
■ Implicit / Explicit / Hierarchical / Specialized End-to-End VLA

OpenVLA



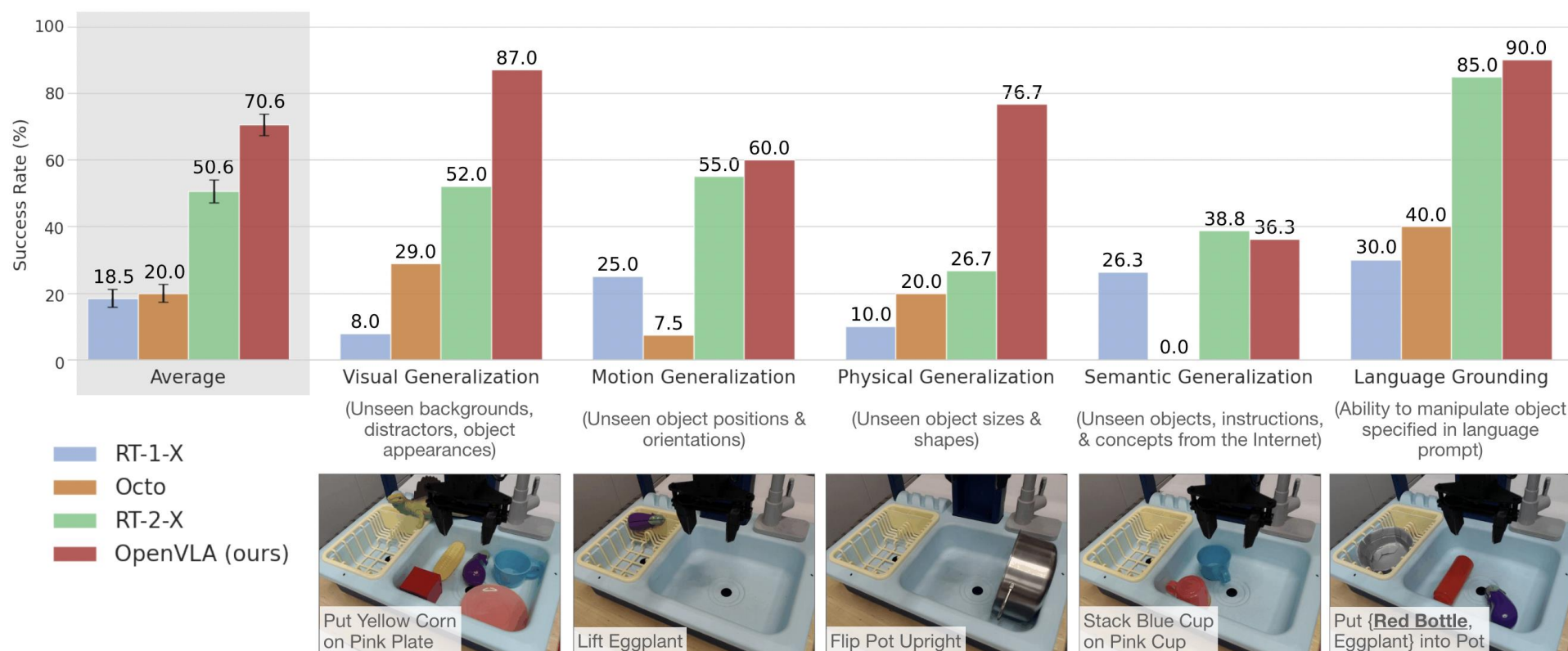
■ Implicit / Explicit / Hierarchical / Specialized End-to-End VLA

OpenVLA — Framework



■ Implicit / Explicit / Hierarchical / Specialized End-to-End VLA

OpenVLA — Results



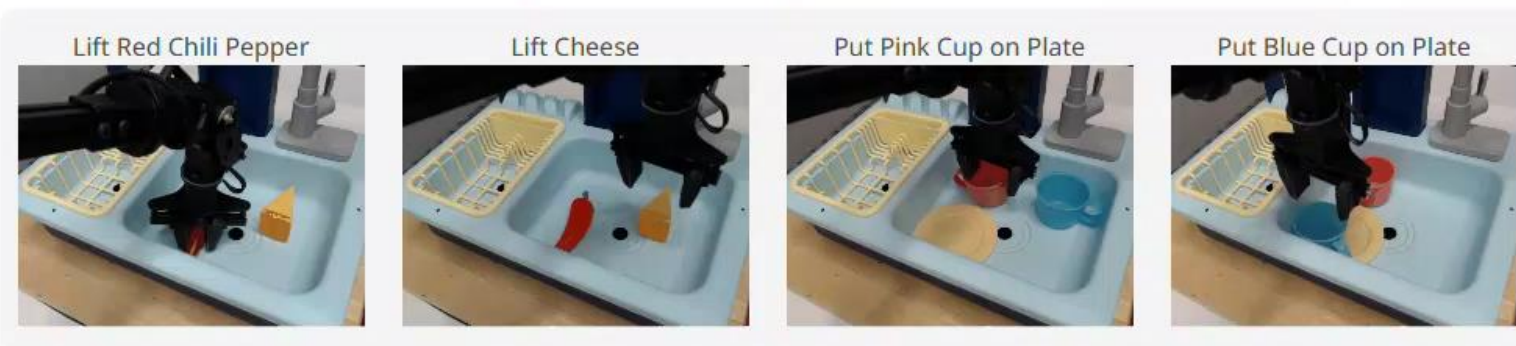
■ Implicit / Explicit / Hierarchical / Specialized End-to-End VLA

OpenVLA — Results

In a scene with many distractor objects, OpenVLA is able to approach and manipulate the correct target object.



Similarly, we assess the policy's language grounding by prompting it to manipulate different target objects given the same initial states. We find that OpenVLA reliably targets the correct object in most cases.



■ **Implicit** / Explicit / Hierarchical / Specialized **End-to-End VLA**

RT-2 vs OpenVLA

RT-2

55B Parameters

VIT + PaLM

Highlight: Semantic Reasoning

OpenVLA

7B Parameters

DINOv2 + SigLIP + Llama-7B

Highlight: Quick Fine-Tuning

■ Implicit / Explicit / Hierarchical / Specialized End-to-End VLA

RT-2 vs OpenVLA

In-Distribution & Basic Out-of-Distribution tasks

RT-2-X:
Pick Coke Can
(In-Distribution)



OpenVLA:
Pick Coke Can
(In-Distribution)



RT-2-X:
Move Banana near Plate
(OOD: unseen target object)



OpenVLA:
Move Banana near Plate
(OOD: unseen target object)



RT-2-X:
Place Banana on Plate
(OOD: unseen target object & instruction)



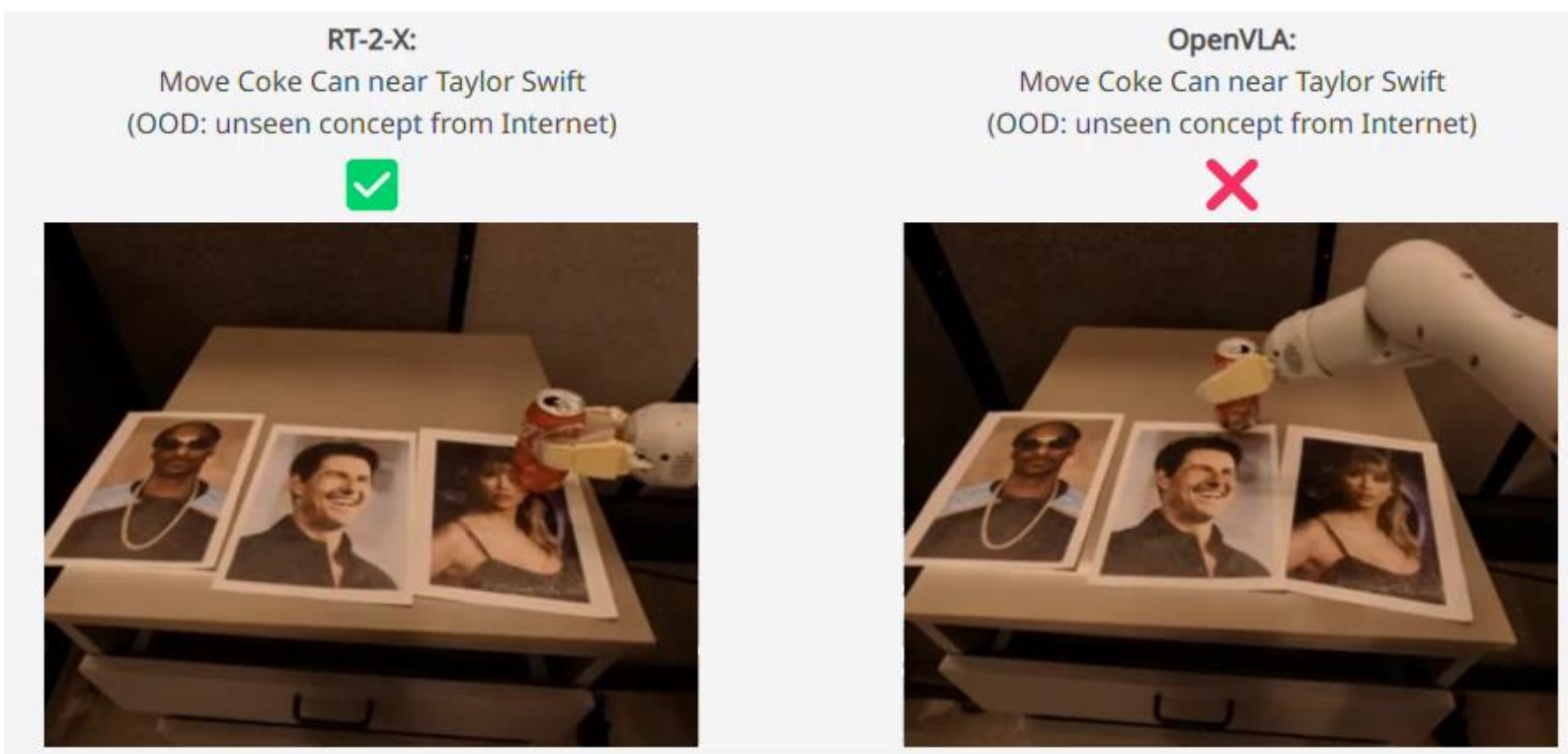
OpenVLA:
Place Banana on Plate
(OOD: unseen target object & instruction)



■ Implicit / Explicit / Hierarchical / Specialized End-to-End VLA

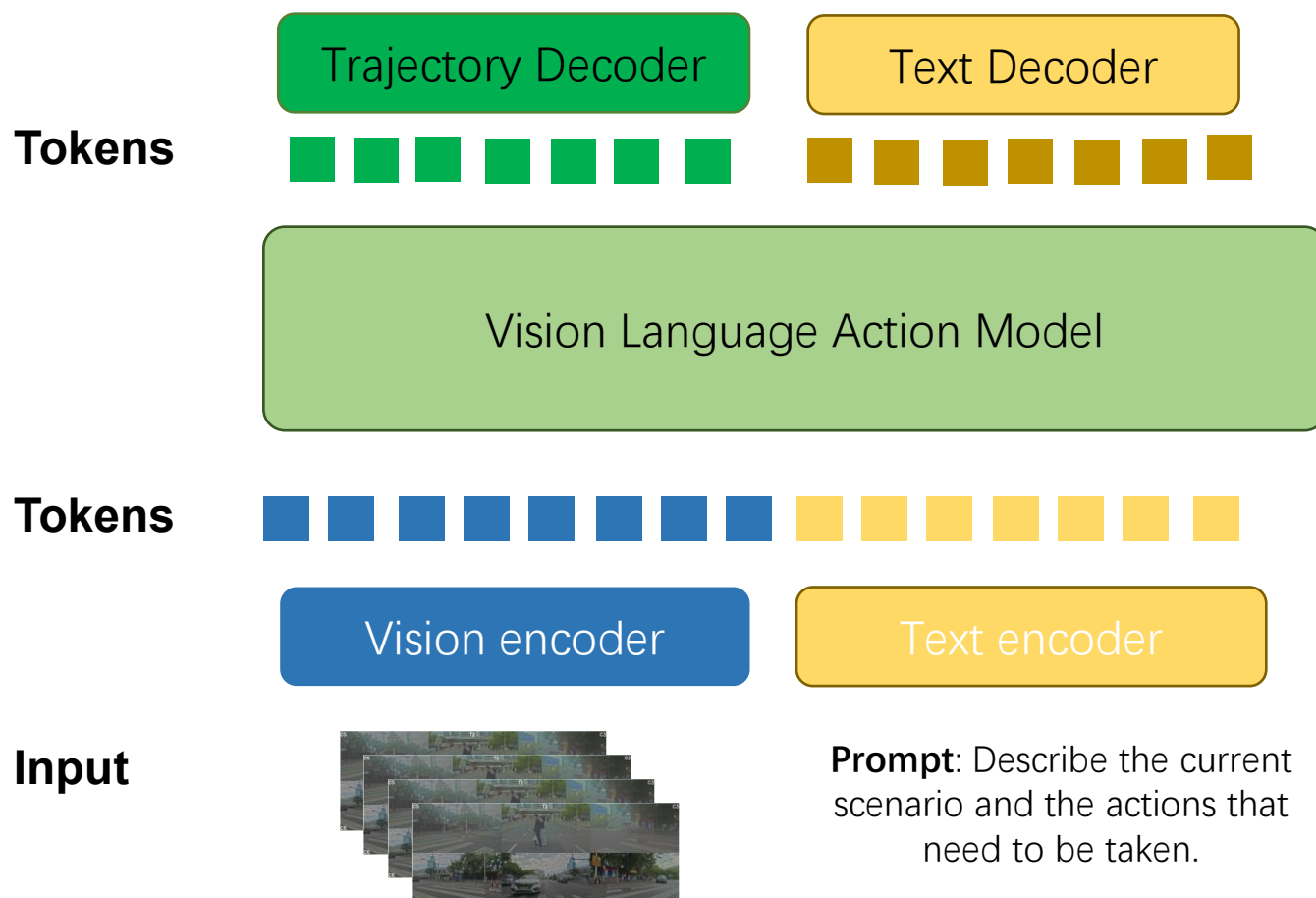
RT-2 vs OpenVLA

Out-of-Distribution Concepts



■ Implicit / Explicit / Specialized End-to-End VLA

VLA Model for Autonomous Driving



Output

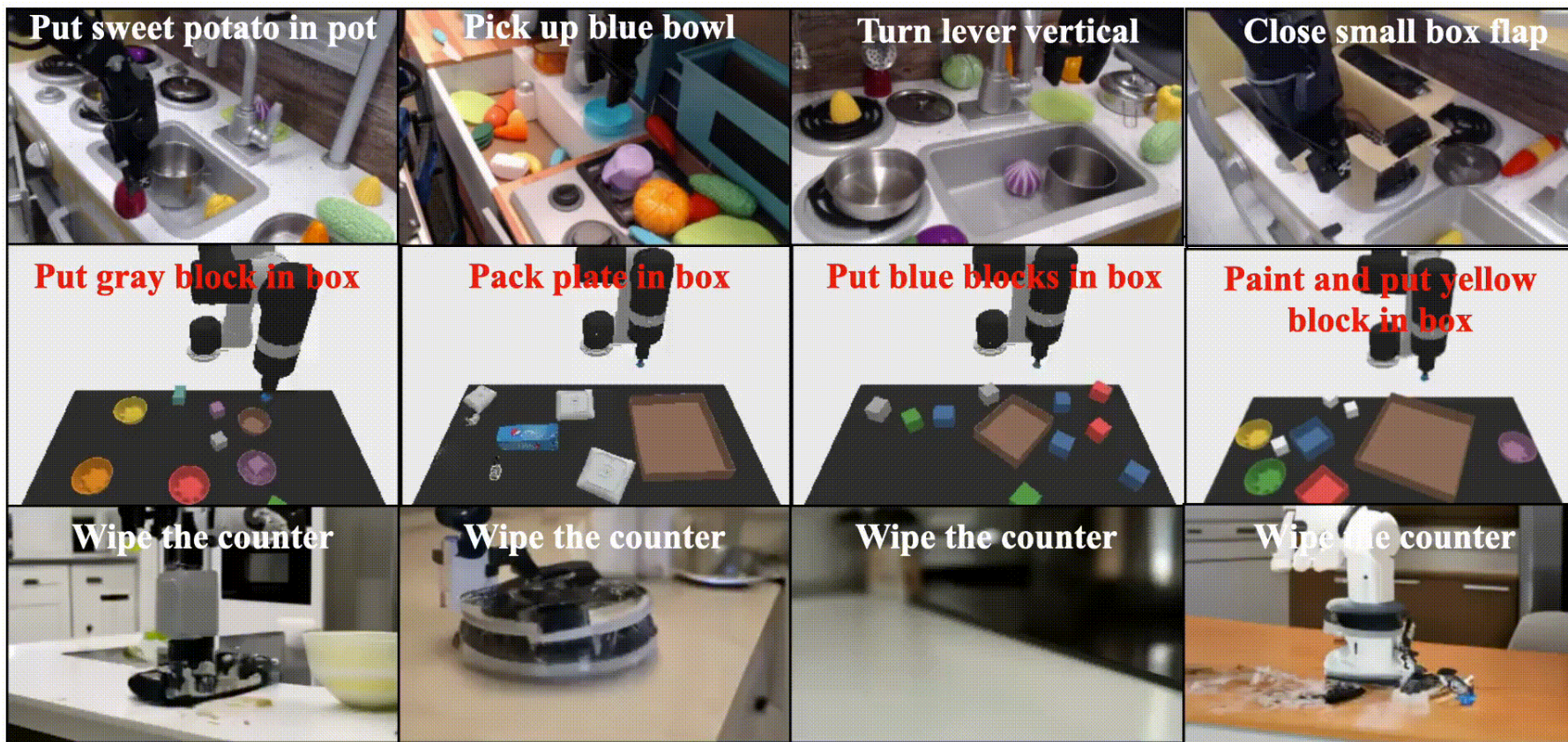


Two pedestrians are crossing the street, need to slow down and wait for them to pass.

- **Scalable:** Tons of data for scaling law
- **General:** Techniques like LoRA, RLHF, DPO are directly applicable

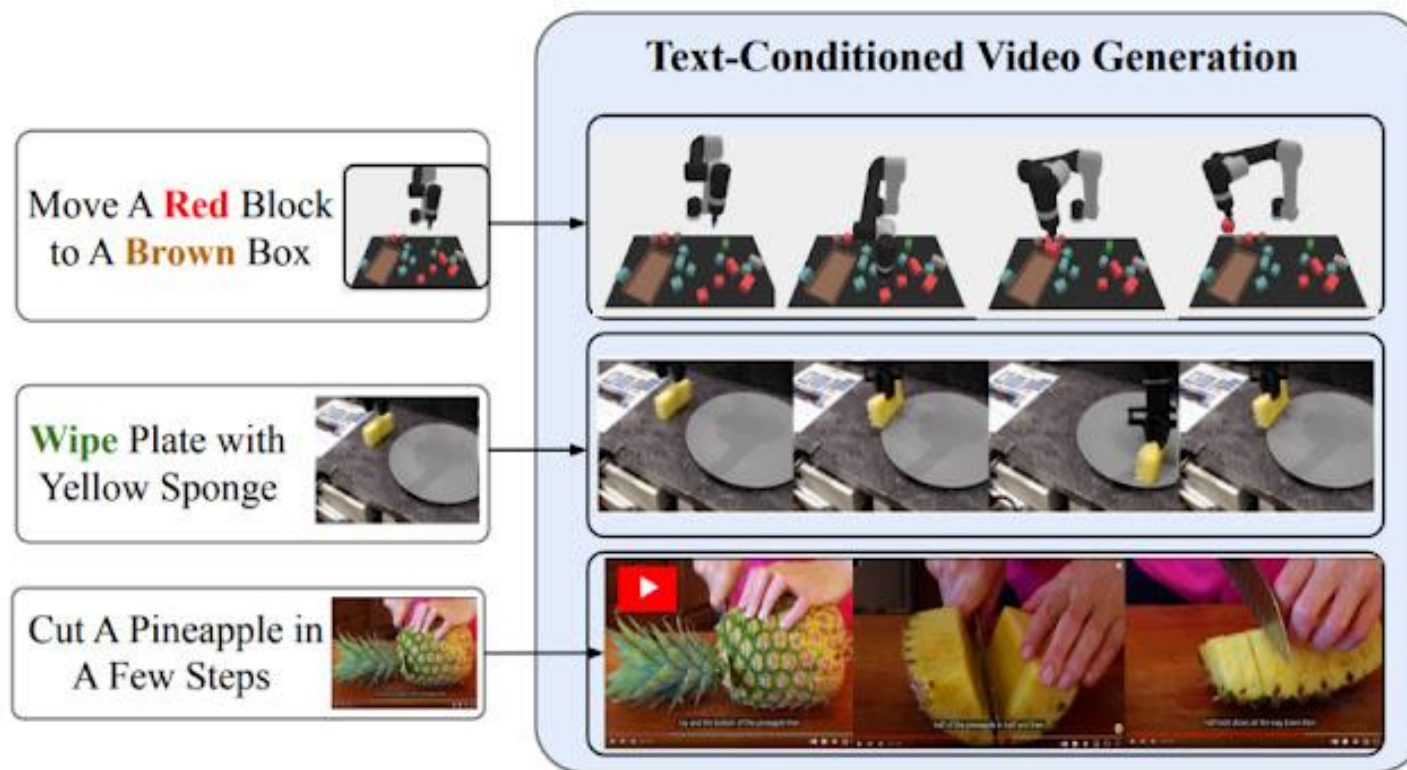
■ Implicit / **Explicit** / Hierarchical / Specialized **End-to-End VLA**

UniPi



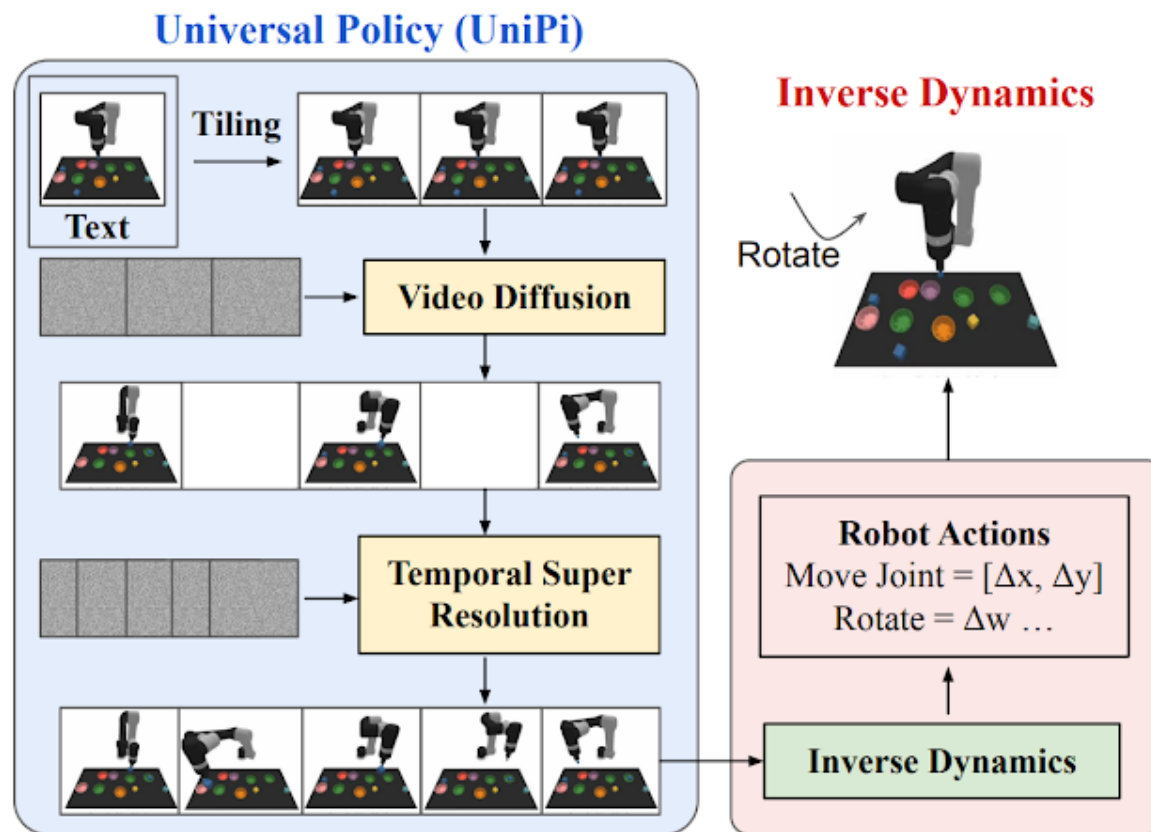
■ Implicit / **Explicit** / Hierarchical / Specialized **End-to-End VLA**

UniPi Represent actions prediction as **video generation task**.



■ Implicit / **Explicit** / Hierarchical / Specialized **End-to-End VLA**

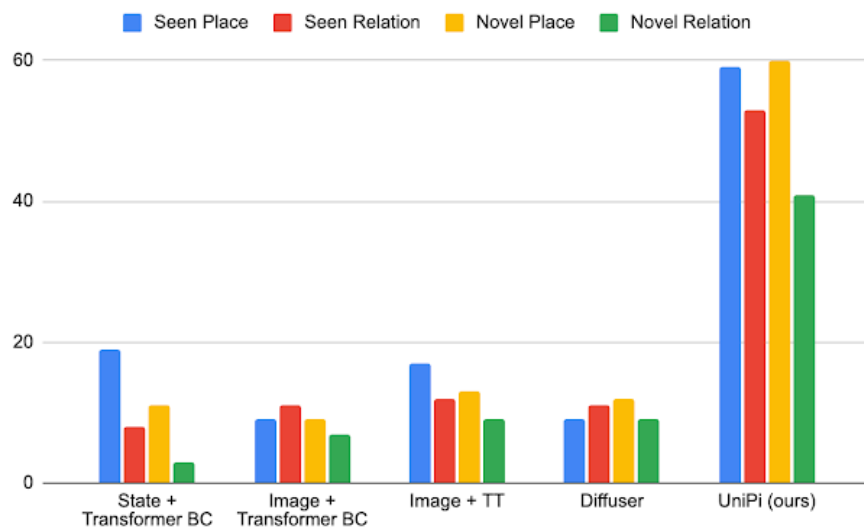
UniPi — Framework



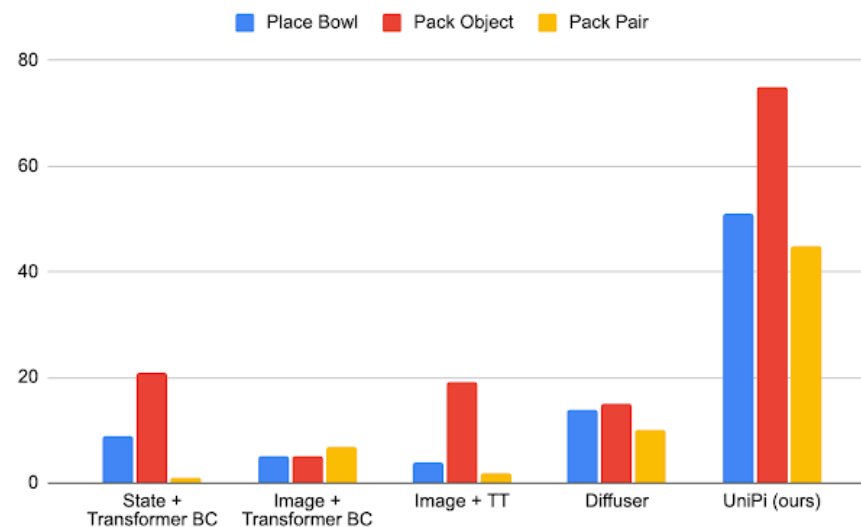
■ Implicit / **Explicit** / Hierarchical / Specialized **End-to-End VLA**

UniPi — Results

Evaluation on novel language-based goals



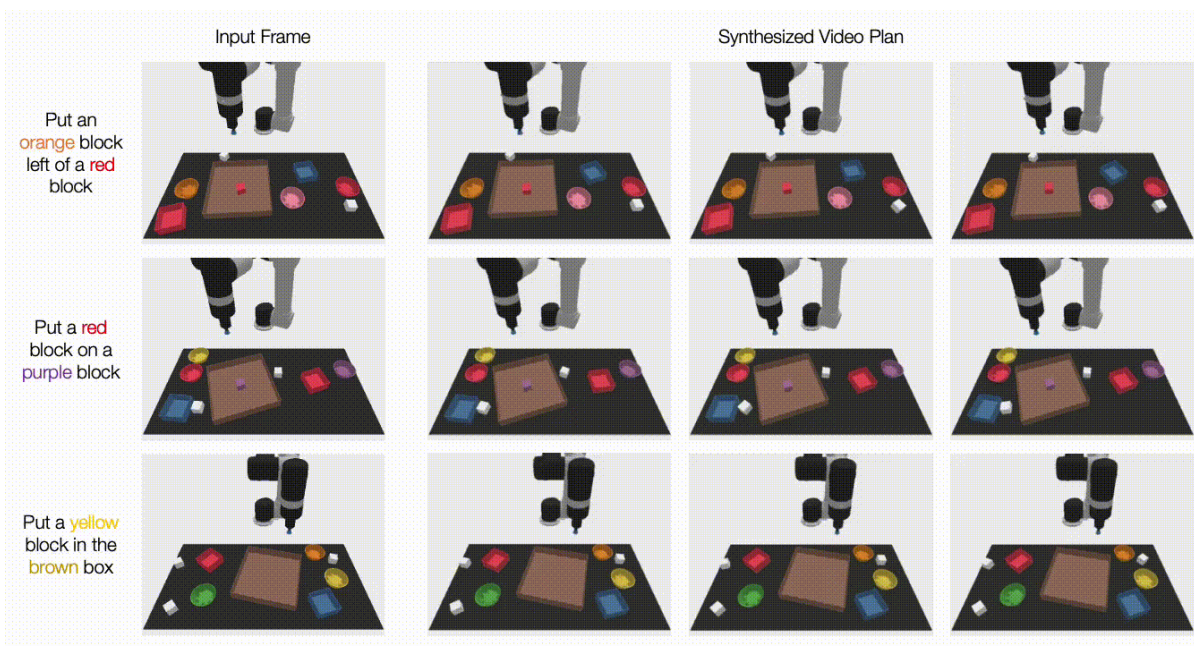
Evaluation on novel tasks



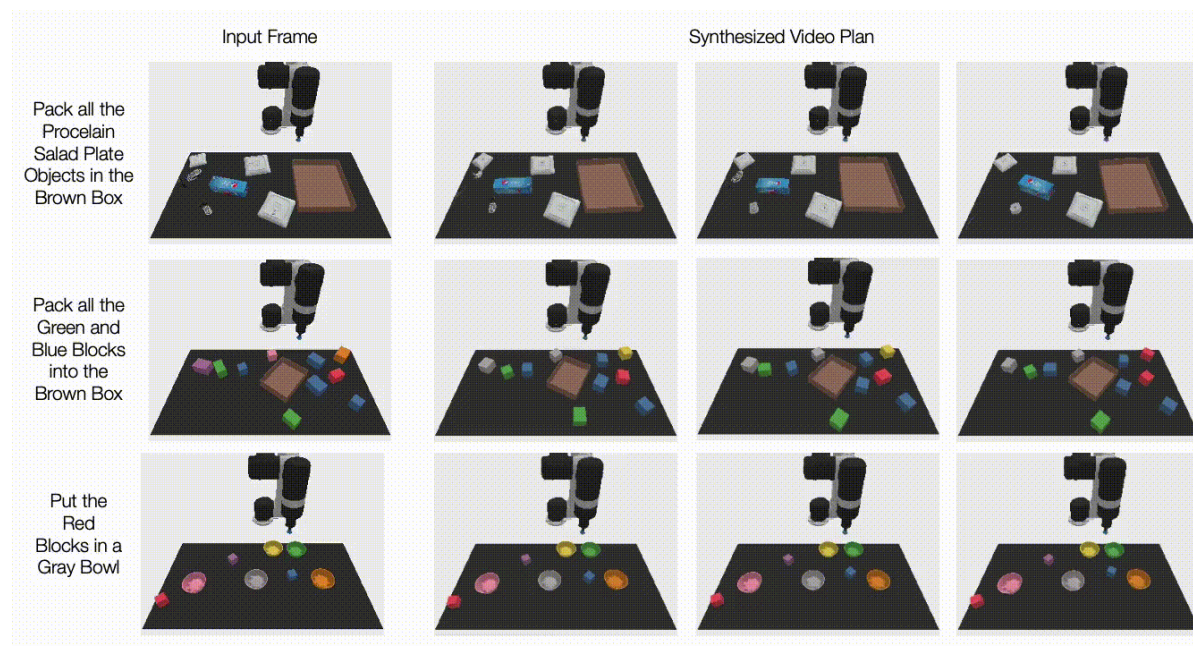
■ Implicit / **Explicit** / Hierarchical / Specialized **End-to-End VLA**

UniPi — Results

Evaluation on novel language-based goals



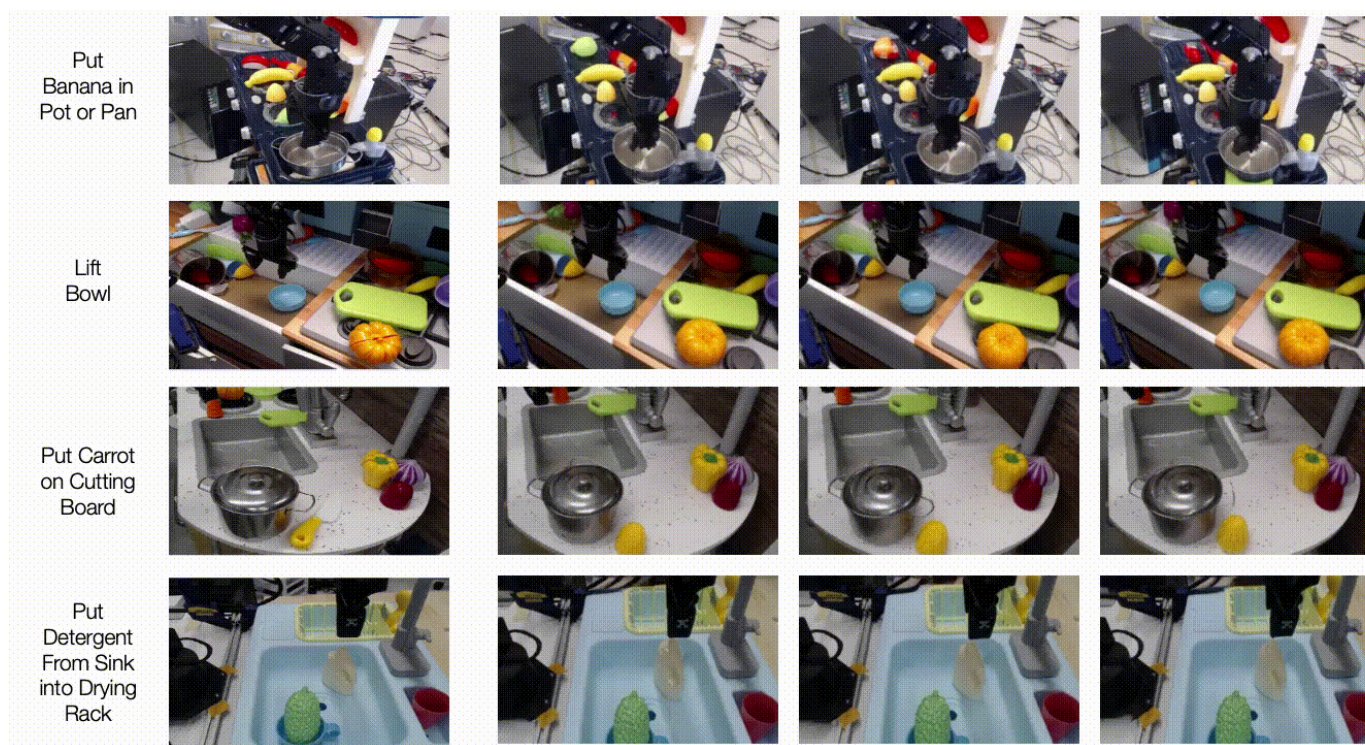
Evaluation on novel tasks



■ Implicit / **Explicit** / Hierarchical / Specialized **End-to-End VLA**

UniPi — Results

Real world transfer



■ Implicit / Explicit / **Hierarchical** / Specialized **End-to-End VLA**

Latent Codes as Bridges (LCB) in Hierarchical Robot Control

Language Table

There is a block closest to the top right. Push that block to the block of the same color

Sort the red blocks into the bottom right corner

Sort red blocks into the bottom left corner

Find the block closest to the bottom center, push it to the block with the same shape.

CALVIN

Open the drawer

Rotate the red cube to the right

t

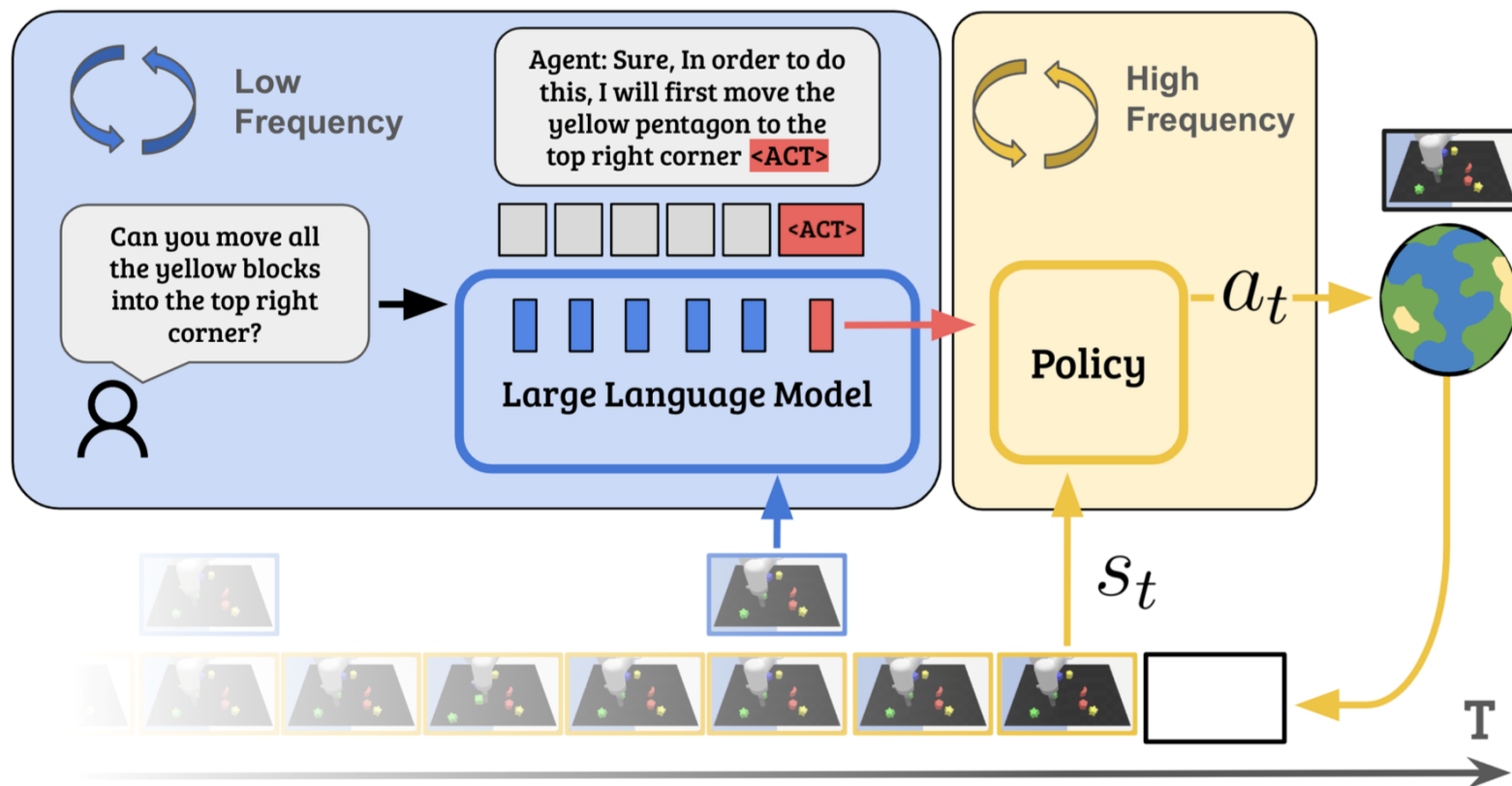
0: lift_pink_block_slider

CALVIN long horizon

■ Implicit / Explicit / **Hierarchical** / Specialized **End-to-End VLA**

LCB — Framework

The feature embedding from the token's last layer serves as a high-level latent goal for the downstream policy network



Latent Codes as Bridges in Hierarchical Robot Control

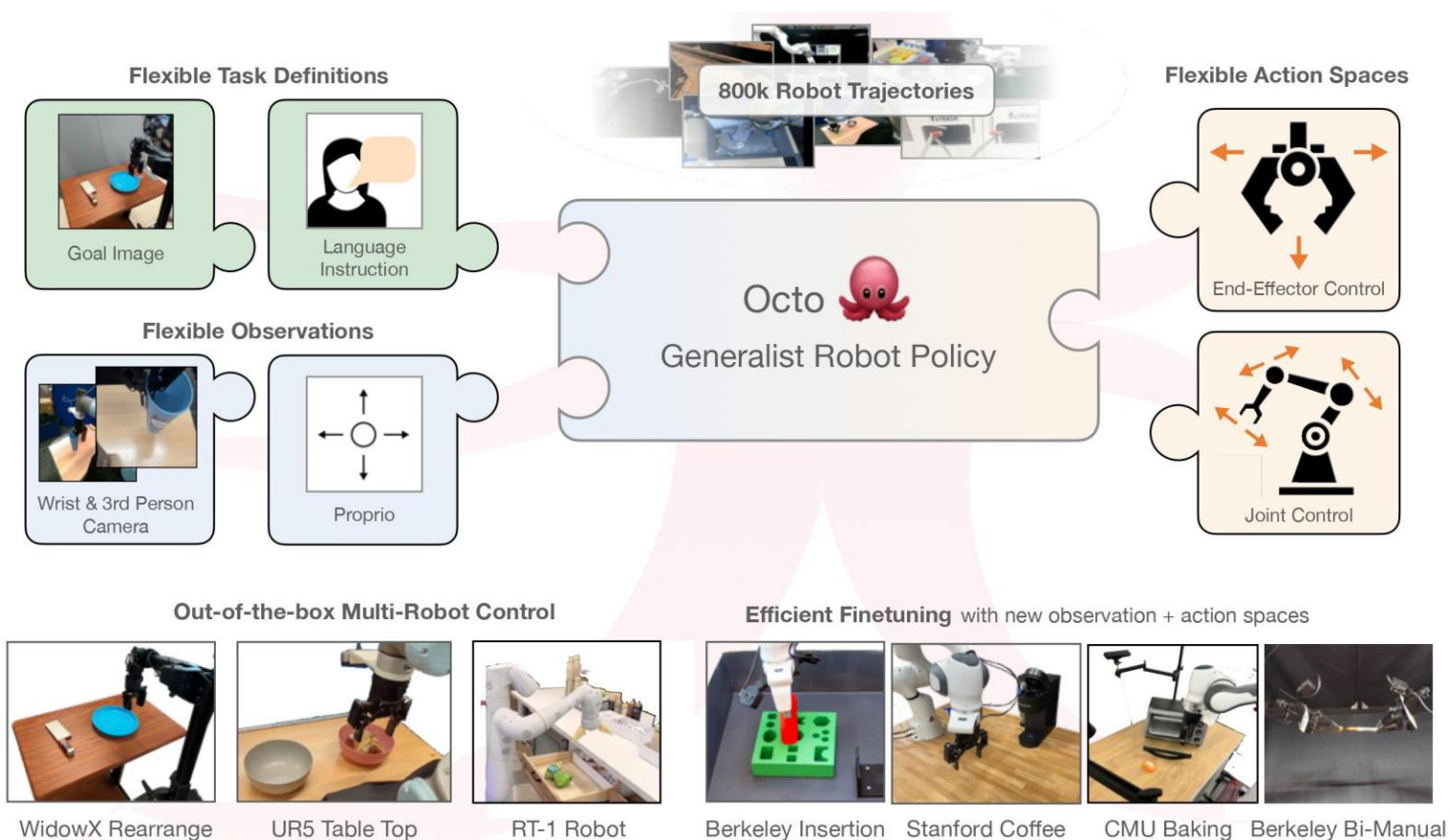
■ Implicit / Explicit / **Hierarchical** / Specialized **End-to-End VLA**

LCB — Results

	Model	RF[33]	3DDA[36]	LCB	
Task Completion rates on CALVIN long horizon task with GPT-4 enriched instruction	Task Completed in a Sequence (Success Rate)	1/5	0.620	0.652	0.736
		2/5	0.330	0.391	0.502
		3/5	0.164	0.203	0.285
		4/5	0.086	0.117	0.160
		5/5	0.046	0.061	0.099
	Avg Len		0.40	1.42	1.78

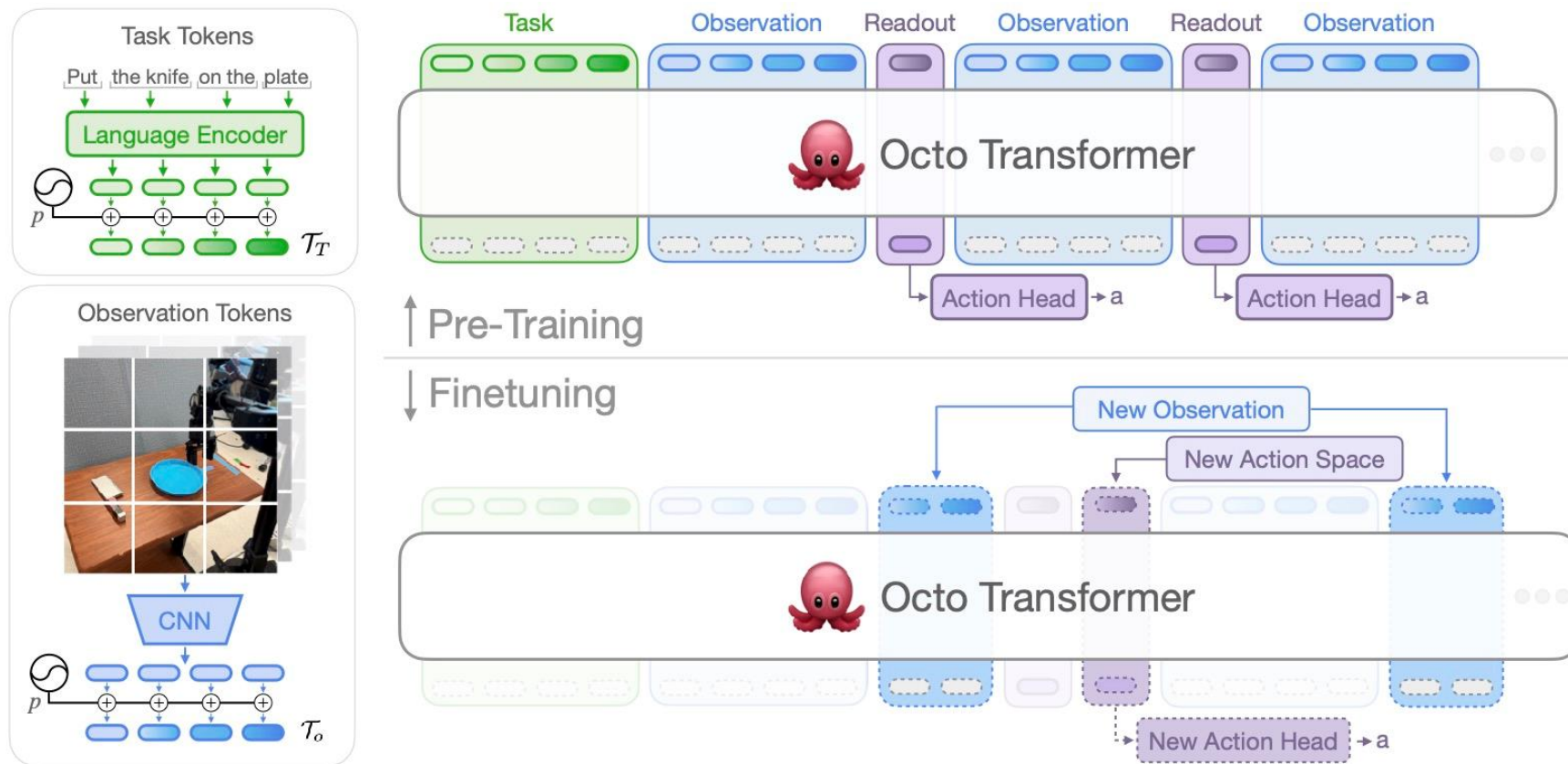
■ Implicit / Explicit / Hierarchical / **Specialized End-to-End VLA**

Octo aims to support a variety of commonly used robots, sensor configurations, and actions



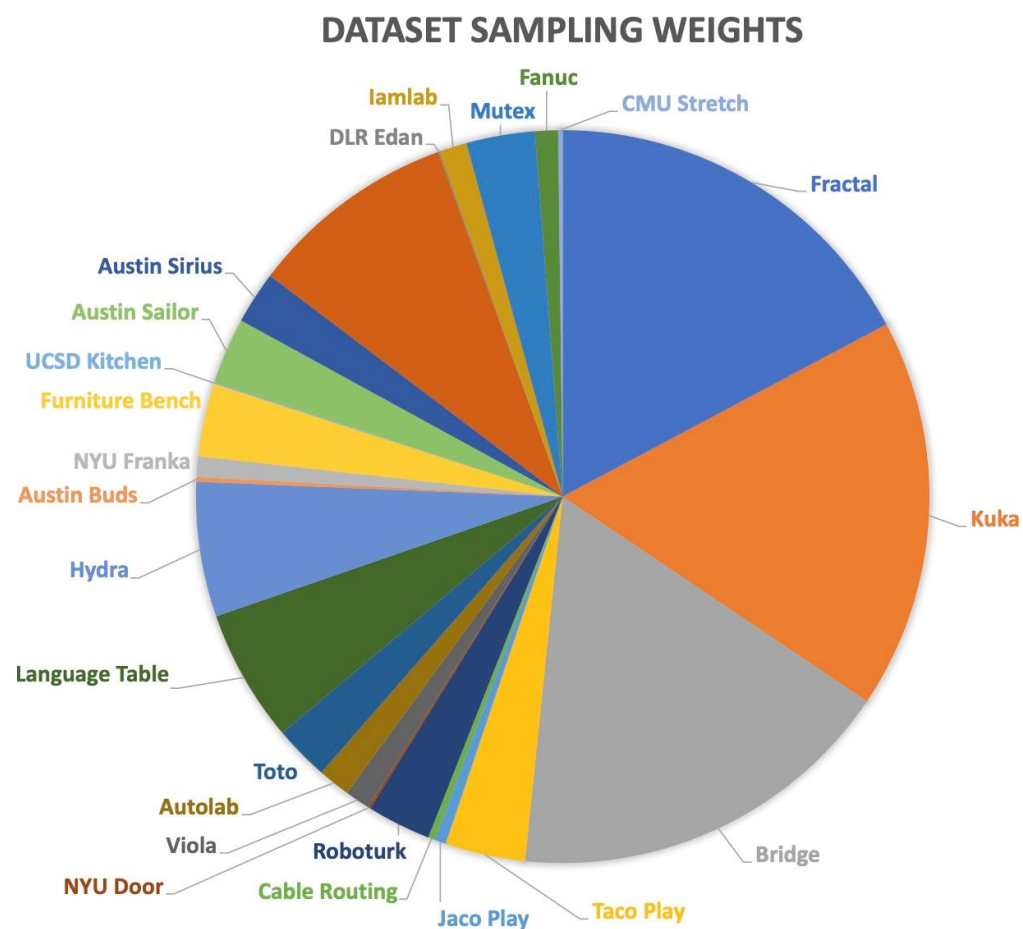
■ Implicit / Explicit / Hierarchical / **Specialized End-to-End VLA**

Octo — Framework



■ Implicit / Explicit / Hierarchical / **Specialized End-to-End VLA**

Octo — Dataset



■ Implicit / Explicit / Hierarchical / **Specialized End-to-End VLA**

Octo — Results

Evaluate Octo on 9 real robot setups

WidowX BridgeV2



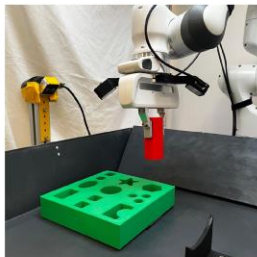
UR5 Tabletop



RT-1 Robot



Berkeley Peg Insert



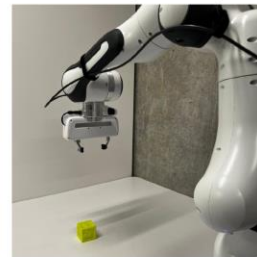
Stanford Coffee



CMU Baking



Berkeley Pick-up



Berkeley Bi-Manual

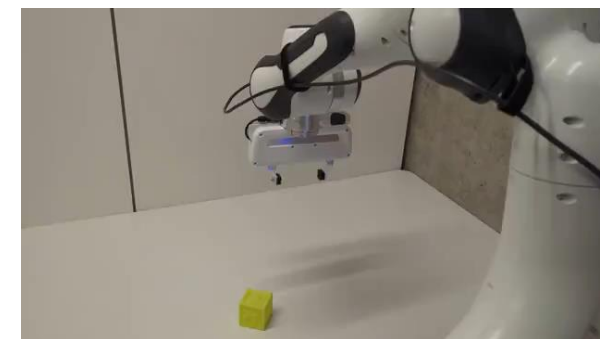
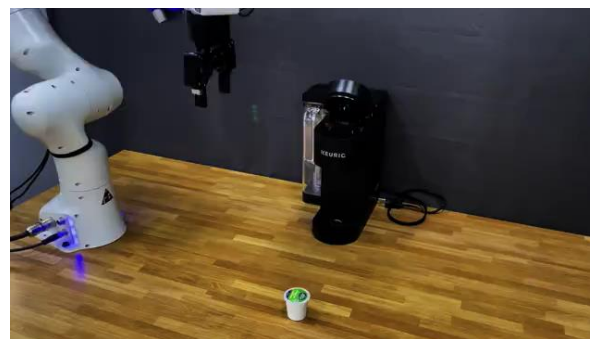
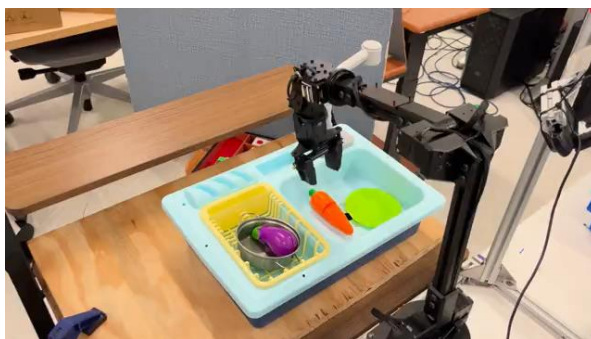


Berkeley Coke



0-Shot Evaluation

Finetuning Evaluation



■ Implicit / Explicit / Hierarchical / Specialized End-to-End VLA

Octo — Results

Zero-shot

	WidowX	UR5	RT-1 Robot
RT-1-X	0.20	0.35	0.60
RT-2-X	0.50	—	0.85
Octo 🐙	0.50	0.70	0.80

Finetuning

	CMU Baking	Stanford Coffee	Berkeley Peg Insert [*]	Berkeley Pick-Up [†]	Berkeley Bimanual [†]	Berkeley Coke	Average
From Scratch	0.25	0.45	0.10	0.00	0.20	0.20	0.20
VC-1	0.30	0.00	0.05	0.00	0.50	0.10	0.15
Octo 🐙	0.50	0.75	0.70	0.60	0.80	1.00	0.72

^{*}New observation input (force-torque proprioception)

[†]New action space (joint position control)

3

VLA Discussion

- **Challenges & Future Direction**
- **Conclusion**

■ Challenges & Future Direction / Conclusion

Scarcity of Robotic Data

Motion Planning

Real-Time Responsiveness

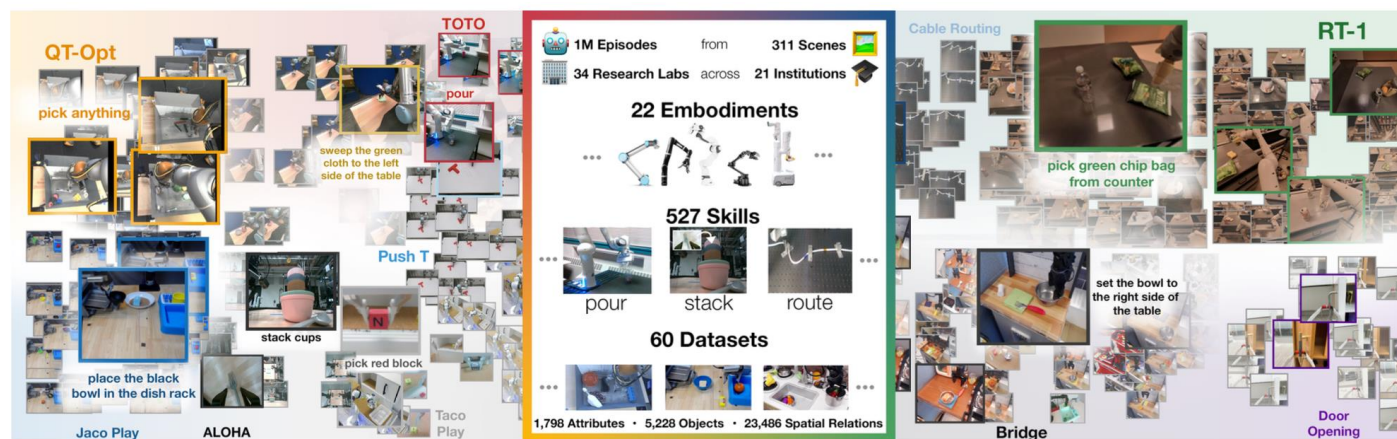
Integration of Multiple Modalities

Generalization to Unseen Scenarios

Long-Horizon Task Execution

Multi-agent Systems

Safety Considerations



Open X-Embodiment suffers from **inconsistent data collection standards** and **outdated robot configurations**.

DROID dataset contains **low-quality data** that can confuse robots during learning.

Large-scale, high-quality, new robotics data is urgently needed !

■ Challenges & Future Direction / Conclusion

Scarcity of Robotic Data

Motion Planning

Real-Time Responsiveness

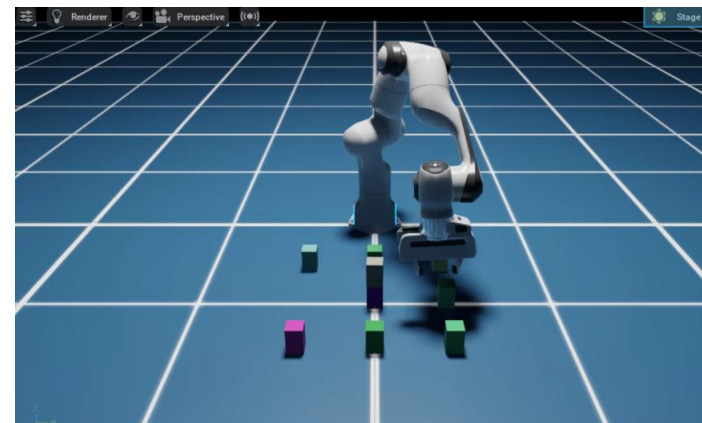
Integration of Multiple Modalities

Generalization to Unseen Scenarios

Long-Horizon Task Execution

Multi-agent Systems

Safety Considerations



Current motion planning modules **struggle with complex environments**, limiting robots' ability to use tools, navigate, and perform precise tasks.

More robust algorithms are needed to overcome these challenges !

■ Challenges & Future Direction / Conclusion

Scarcity of Robotic Data

Motion Planning

Real-Time Responsiveness

Integration of Multiple Modalities

Generalization to Unseen Scenarios

Long-Horizon Task Execution

Multi-agent Systems

Safety Considerations

Robotic applications **need real-time decisions and actions**.

VLA models should be responsive with **low latency**.

The entire robot system, from task planning to motion planning,
requires global optimization.

■ Challenges & Future Direction / Conclusion

Scarcity of Robotic Data

Motion Planning

Real-Time Responsiveness

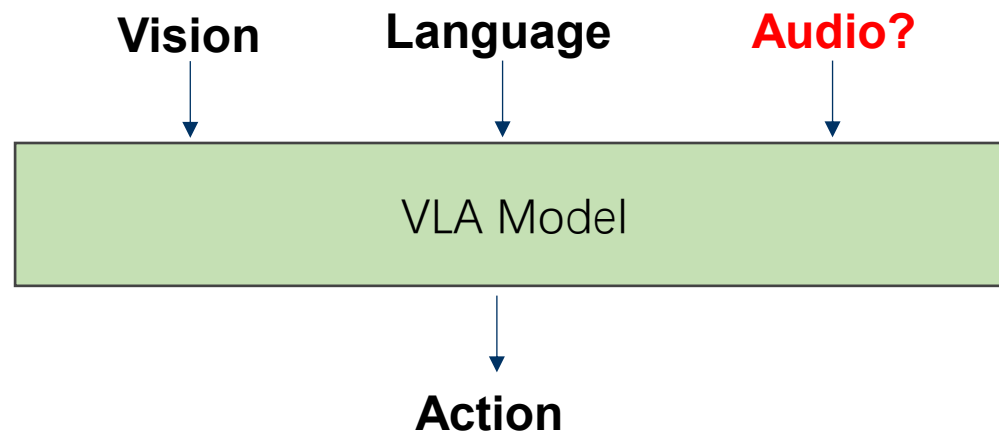
Integration of Multiple Modalities

Generalization to Unseen Scenarios

Long-Horizon Task Execution

Multi-agent Systems

Safety Considerations



Improving multi-modal learning, fusion, and adaptation is necessary.

Moreover, **adding audio and speech** can enhance robots' collaboration with users.

■ Challenges & Future Direction / Conclusion

Scarcity of Robotic Data

Motion Planning

Real-Time Responsiveness

Integration of Multiple Modalities

Generalization to Unseen Scenarios

Long-Horizon Task Execution

Multi-agent Systems

Safety Considerations

A versatile robot should understand and execute natural language instructions in various scenarios. Achieving this requires **robustness to changes in instructions, environments, and objects**, similar to ChatGPT's language capabilities.

This calls for adaptable and scalable VLA architectures !

■ Challenges & Future Direction / Conclusion

Scarcity of Robotic Data

Motion Planning

Real-Time Responsiveness

Integration of Multiple Modalities

Generalization to Unseen Scenarios

Long-Horizon Task Execution

Multi-agent Systems

Safety Considerations

Single instructions like "clean the room" involve complex tasks requiring robots to plan and execute multiple actions over time. Current planners struggle because LLMs aren't tuned for embodied tasks.

Efficient planners with strong perception and common sense are needed!

■ Challenges & Future Direction / Conclusion

Scarcity of Robotic Data

Motion Planning

Real-Time Responsiveness

Integration of Multiple Modalities

Generalization to Unseen Scenarios

Long-Horizon Task Execution

Multi-agent Systems

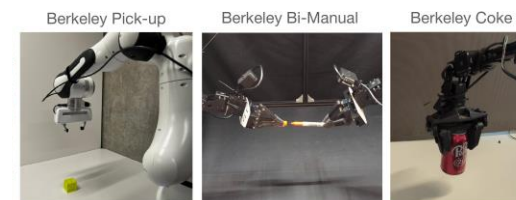
Safety Considerations

Deploying many robots is desirable but challenging due to multi-agent cooperation issues. Effective communication and task distribution are difficult, especially with diverse robot fleets.

While multi-agent systems offer benefits like distributed perception and collaborative fault recovery, fully realizing these advantages is still challenging.



...



■ Challenges & Future Direction / Conclusion

Scarcity of Robotic Data

Motion Planning

Real-Time Responsiveness

Integration of Multiple Modalities

Generalization to Unseen Scenarios

Long-Horizon Task Execution

Multi-agent Systems

Safety Considerations



Safety is crucial in robotics due to real-world interactions. Ensuring safety involves integrating commonsense reasoning, robust safety mechanisms, risk assessments, and human-robot interaction protocols. Additionally, making VLA decision-making interpretable and expandable is vital for diagnosing errors and troubleshooting.

■ Challenges & Future Direction / Conclusion

VLA promises significant advancements in enabling robots to interact with their surroundings.

Recent progress has shown these models can complete complex tasks under diverse conditions.

However, **challenges in generalization, efficiency, and safety** remain.

Further research is essential to overcome these obstacles and enable the **widespread adoption** of

VLA-powered robots in real-world applications.